

Research Note on MathOverflow 413935

Exact relaxations, finite-temperature barriers, and a sharper proof frontier

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Status and nonclaim. This note does not prove that the limit in Question 1 exists, and it does not prove nonexistence. It contains independently checked partial theorems and exact finite computations. It must not be cited or submitted as a solution to the MathOverflow question.

Abstract

For a symmetric zero-diagonal sign matrix A , put

$$M(A) = \max_{x \in \{-1,1\}^n} \left| \frac{1}{2} x^\top A x \right|, \quad F(n) = \min_A M(A).$$

The open question is whether $F(n)/n^{3/2}$ converges. We first audit the previous bounds $1/\pi \leq \liminf \leq \limsup \leq 1/2$, the augmented cut-code identity, and one-vertex monotonicity; all survive audit. We prove that the problem is equivalent up to an explicit $O(n)$ error to minimizing density-1/2 cut discrepancy over graphs with $\lfloor \binom{n}{2}/2 \rfloor$ edges. We then show that the proposed cube relaxation has the unique optimizer zero at zero temperature and at every fixed unscaled positive temperature, with a leading-order zero-temperature integrality gap. In contrast, an ellipsope relaxation has the exact thermodynamic limit 1/2, while the defined two-covariance, edgewise-oriented Gaussian certificate has asymptotic lower-bound constant $1/\pi$. We identify the augmented cut code as a binary linear code whose dual consists of even-cardinality Eulerian subgraphs, and derive an exact signed-Eulerian MacWilliams formula. Ordinary graphons and limiting empirical spectra are proved to erase the $n^{3/2}$ functional. Exact computer-assisted computation gives $F(2), \dots, F(14) = 1, 3, 4, 4, 5, 9, 10, 12, 13, 17, 18, 20, 21$. Exact cross-block computation shows that optimal blocks of orders 2 and 8 cannot occur together in an order-10 optimum. We also identify the one-vertex Bellman state as an exact energy-weighted projective covering radius and certify that the code of exact maximizers, and even the complete scalar partition-function curve, can be insufficient. Retaining the nonlinear arcsine term gives a trace-four bound; a parity theorem for its Gram defect yields $F(20) \geq 30$ and $F(21) \geq 32$. The fully optimized one-parameter certificate nevertheless has exact asymptotic constant $1/\pi$. We derive an exact multivertex weighted Bellman identity, then prove that its weighted-entropy union-bound target has an unavoidable leading constant gap. The replacement is an exact switching orbit noncoverage problem on the projective rank-one group. A complete order-nine catalogue shows that even the energy histogram plus the two-point overlap law of exact maximizers does not determine extension, while the order-ten catalogue exhibits three exact finite-temperature optimizer phases. The annealed-normalized negative replica is exactly supermultiplicative, but the natural power-saving scalar transport conjecture is false with a rigorous n^2 -scale gap. We derive an exact entropy-production identity and stability remainder for that gap. The surviving block law has a nonnegative conditional relative-switching alignment term whose zero-temperature slope is exactly the joint optimizer-composition gain. Its first Hamiltonian is an explicit mixed four-cycle trace, and exact examples show that all three local scalar partition curves fail to

determine it. A new deterministic microcanonical theorem nevertheless extracts a guaranteed composition gain: the relevant quantile is the 2^{n+k-1} -st smallest sum of the three local energy deficits. This is optimal after discarding the additive labels and turns near-maximal-state thinness into a precise large-deviation target. An equivalent cumulant criterion is exact, but every quadratic-cumulant majorant has normalized certificate floor $\sqrt{\log 2/2} > 1/2$; ordinary degree-two hypercontractivity also misses the required exponential tail scale. Exact balanced-split calibration through order fourteen exhibits a substantial finite additive-geometry bonus without closing the scalar route asymptotically. Retaining the additive labels gives an exact Parseval refinement and a complete integer-moment vacancy hierarchy. A degree-19 certificate detects the unique empty fiber in the tested balanced C_{14} shell, while a matching abstract moment pair proves that bounded-order moments cannot be uniformly sufficient without signing-specific shell structure. Exact fixed-density and equal-cloud estimates also close separate cross-norm control and ordinary complete/empty blow-up repair, but leave joint labeled cancellation open. We also record the known regular Paley construction: square-order Paley conference matrices have Boolean eigenvectors attaining their spectral ceiling, a statement about those matrices rather than the minimax F . For all symmetric Paley orders we derive an exact Fourier-leakage identity: spectral saturation is equivalent to an asymptotically balanced Boolean function with asymptotically one-sided quadratic-character frequency energy. An interval construction controlled by the least quadratic nonresidue proves this condition on a multiplicatively dense sequence of prime orders. For arbitrary conference matrices we identify spectral alignment exactly with maximal ℓ_1 norm in the two eigenspaces. The remaining dense Paley minimax-rigidity estimate is equivalent, not merely sufficient, to the full limit having value $1/2$. Exhaustive source-built evaluation gives $M(C_{30}) = 75$, but a separate minimax rigidity lower bound would still be needed to turn this construction into a statement about the limit of F . Finally, an explicit scalar countermodel shows that even uniform $O(\sqrt{n})$ Bellman control does not force convergence. The sharp remaining target is a quenched minimax free-energy limit, or a signing-specific degree- $O(N)$ negative localizing form derived from the factored labeled-shell coefficients with a summable composition error.

1 Problem, audited baseline, and scope

The question, asked by Paata Ivanisvili on MathOverflow in 2022 [1], is whether

$$\lim_{n \rightarrow \infty} \frac{F(n)}{n^{3/2}}$$

exists. Here $A_{ii} = 0$, $A_{ij} = A_{ji} = a_{ij} \in \{-1, 1\}$, and

$$Q_A(x) = \sum_{i < j} a_{ij} x_i x_j = \frac{1}{2} x^\top A x.$$

The earlier attempt was rechecked from source rather than assumed. Its central conclusions are correct:

$$\frac{1}{\pi} \leq \liminf_{n \rightarrow \infty} \frac{F(n)}{n^{3/2}} \leq \limsup_{n \rightarrow \infty} \frac{F(n)}{n^{3/2}} \leq \frac{1}{2}, \quad (1)$$

$$F(n) = \binom{n}{2} - 2\rho(D_n), \quad F(n) \leq F(n+1) \leq F(n) + n. \quad (2)$$

The lower bound uses two Gaussian sign roundings and the arcsine law. The upper bound uses symmetric Paley conference matrices and principal submatrices. The code D_n is the cut-character set augmented by its negative. A claim-by-claim audit, including two attribution errors and a factor-of-two prose correction, is in `AUDIT.md`. Nothing in (1) or (2) implies convergence.

All new statements below distinguish a proved identity from a proposed limit theorem. Exact computations guide the search but are not used as asymptotic proofs.

2 An exact dense cut-discrepancy reformulation

The sign problem is, up to an additive linear term, precisely the extremal cut-discrepancy problem for graphs having $\lfloor \binom{n}{2}/2 \rfloor$ edges. Put $m = \binom{n}{2}$. For a graph G on $[n]$, define

$$\text{cdisc}_{1/2}(G) = \max_{S \subseteq [n]} \left| e_G(S, S^c) - \frac{1}{2}|S||S^c| \right|,$$

and let

$$H(n) = \min \left\{ \text{cdisc}_{1/2}(G) : e(G) = \left\lfloor \frac{m}{2} \right\rfloor \right\}.$$

Theorem 1 (Fixed-density cut-discrepancy equivalence). *For every $n \geq 2$,*

$$-1 \leq 4H(n) - F(n) \leq \sqrt{\binom{n}{2}} + 2. \quad (3)$$

Consequently, $F(n)/n^{3/2}$ converges if and only if $4H(n)/n^{3/2}$ converges, and the two limits are equal.

Proof. For a signing A , let G_A contain the negative edges and put $t = \sum_{i < j} a_{ij} = m - 2e(G_A)$. If x_S is positive on S and negative on its complement, and $k = |S|(n - |S|)$, then

$$Q_A(x_S) = t - 2k + 4e_{G_A}(S, S^c). \quad (4)$$

Let $t_0 = 0$ when m is even and $t_0 = 1$ when m is odd. On the layer $e(G) = \lfloor m/2 \rfloor$, equation (4) gives

$$e_G(S, S^c) - \frac{k}{2} = \frac{Q_A(x_S) - t_0}{4}.$$

The vectors x_S exhaust the Boolean cube modulo global sign, so

$$|4 \text{cdisc}_{1/2}(G_A) - M(A)| \leq |t_0| \leq 1. \quad (5)$$

It remains to compare the fixed-density layer with all signings. If A minimizes M , switching by a uniformly random $z \in \{-1, 1\}^n$ replaces t by $Q_A(z)$. Walsh orthogonality gives

$$\mathbb{E}_z Q_A(z) = 0, \quad \mathbb{E}_z Q_A(z)^2 = m,$$

so some switch has $|t| \leq \sqrt{m}$. Since $t \equiv t_0 \pmod{2}$, flipping $|t - t_0|/2$ suitably chosen edge signs reaches total sum t_0 . Each edge flip changes every value of Q_A by at most 2, hence the new maximum is at most $F(n) + |t - t_0| \leq F(n) + \sqrt{m} + 1$. Minimizing (5) on the fixed-density layer proves (3). $\square$

Backurs and Bavarian use the density-centered parameter

$$\max_S |e_G(S, S^c) - \rho_G |S||S^c||$$

and prove that it is $\Omega(\min(\rho_G, 1 - \rho_G)n^{3/2})$ for every graph [6]. On the half-density layer this recovers the correct scale but neither a sharp constant nor an asymptotic limit. Earlier graph discrepancy work of Erdős, Goldberg, Pach, and Spencer is closely related, but its balanced bipartite discrepancy is not identical to the all-cuts, absolute parameter above [7].

3 The relaxed-coupling program collapses

Let $m = \binom{n}{2}$, let $c_x = (x_i x_j)_{i < j}$, and for arbitrary real couplings define

$$G_n(a) = \max_{x \in \{-1, 1\}^n} |\langle a, c_x \rangle|.$$

Theorem 2 (Cube relaxation). *The function G_n is a norm on $\mathbb{R}^m$. Hence*

$$\min_{a \in [-1, 1]^m} G_n(a) = 0,$$

and the unique minimizer is $a = 0$.

Proof. Homogeneity and the triangle inequality follow from the maximum of absolute linear forms. If X is uniform on $\{-1, 1\}^n$, orthogonality of distinct degree-two Walsh characters gives

$$\mathbb{E} \langle a, c_X \rangle^2 = \sum_{i < j} a_{ij}^2.$$

Thus $G_n(a) = 0$ forces $a = 0$. Since the origin belongs to the cube, the last assertion follows. $\square$

In particular, any convex feasible set in the original coupling coordinates that contains every sign coupling contains their convex hull, the whole cube, and therefore contains this zero optimizer when the objective remains G_n . By the audited lower bound, its additive integrality gap is at least $n\sqrt{n-1}/\pi$, of the full target order.

The same collapse persists at positive temperature. Retain duplicated states $(x, s) \in \{-1, 1\}^n \times \{-1, 1\}$ and put

$$L_{n,t}(a) = \frac{1}{t} \log \sum_{x,s} \exp(t \langle a, sc_x \rangle), \quad t > 0.$$

Theorem 3 (Finite-temperature cube dual). *The function $L_{n,t}$ is strictly convex and has the unique cube minimizer zero, with value*

$$\min_{a \in [-1, 1]^m} L_{n,t}(a) = \frac{(n+1) \log 2}{t}.$$

It has the exact dual representation

$$\max_{p \in \Delta(\{-1, 1\}^n \times \{-1, 1\})} \left\{ \frac{H(p)}{t} - \|\mathbb{E}_p[sc_x]\|_1 \right\}.$$

Proof. The Hessian of $L_{n,t}$ is t times the covariance matrix of sc_x under a full-support Gibbs distribution. If a linear functional of these vectors were constant, replacing s by $-s$ would make the constant zero; Walsh orthogonality would then make the functional zero. The Hessian is therefore positive definite. Symmetry makes the gradient vanish at zero, where the value is $t^{-1} \log 2^{n+1}$.

The entropy variational formula for log-sum-exp gives

$$L_{n,t}(a) = \max_p \left\{ \langle a, \mathbb{E}_p[sc_x] \rangle + \frac{H(p)}{t} \right\}.$$

Both domains are compact and convex and the displayed function is convex in a and concave in p , so Sion's theorem applies. Minimization of a linear form over the cube is minus the ℓ^1 -norm of its coefficient. $\square$

Thus the simplest probability-measure dual is exact but vacuous for the discrete problem: the uniform state measure cancels every first moment.

There is nevertheless a useful conditional rounding statement.

Proposition 4 (Variance-sensitive sign rounding). *For $a \in [-1, 1]^m$, set $V(a) = \sum_e (1 - a_e^2)$. There is a sign vector $\sigma \in \{-1, 1\}^m$ for which*

$$G_n(\sigma) \leq G_n(a) + \sqrt{2V(a)(n+1)\log 2} + \frac{4}{3}(n+1)\log 2.$$

Proof. Round the coordinates independently with $\mathbb{E}\sigma_e = a_e$. For fixed x , the centered error $\sum_e (\sigma_e - a_e)c_x(e)$ has variance $V(a)$ and summands bounded by two. The two-sided Bernstein bound with parameter u is at most $2e^{-u}$ beyond $\sqrt{2V(a)u} + 4u/3$. There are 2^{n-1} distinct cut characters. Taking $u = (n+1)\log 2$ makes the union failure probability at most $1/2$. For a successful outcome, add this uniform error to $G_n(a)$. $\square$

This gives $o(n^{3/2})$ loss if $V(a) = o(n^2)$, but the cube optimizer has $V(0) = m$. The obvious spherical repairs also fail. On $\|a\|_2 = \sqrt{m}$, Parseval gives $G_n(a) \geq \sqrt{m}$, attained by a one-coordinate vector. Adding $|a_e| \leq 1$ to that sphere forces every $|a_e| = 1$, recovering the original discrete problem exactly.

4 Temperature scaling and the cross-block wall

The functional proposed in the question is

$$\Phi_{n,\beta} = \min_A \frac{1}{\beta n^{3/2}} \log \sum_x e^{\beta |Q_A(x)|}.$$

For every fixed $\beta > 0$, the maximum-versus-log-sum-exp inequality gives

$$\frac{F(n)}{n^{3/2}} \leq \Phi_{n,\beta} \leq \frac{F(n)}{n^{3/2}} + \frac{\log 2}{\beta \sqrt{n}}. \quad (6)$$

Thus existence of $\lim_n \Phi_{n,\beta}$ for even one fixed positive β is equivalent to the original limit. At this normalization, fixed temperature is already asymptotically zero temperature.

A genuinely extensive mean-field functional is

$$\Psi_{n,\beta} = \min_A \frac{1}{\beta n} \log \sum_{x,s} \exp \left(\frac{\beta s Q_A(x)}{\sqrt{n}} \right).$$

It satisfies

$$\frac{F(n)}{n^{3/2}} \leq \Psi_{n,\beta} \leq \frac{F(n)}{n^{3/2}} + \frac{(n+1)\log 2}{\beta n}. \quad (7)$$

Consequently, existence of $\lim_n \Psi_{n,\beta}$ for arbitrarily large fixed β would settle the problem: the limsup-minus-liminf gap would be at most $\log 2/\beta$ for every such β .

The direct interpolation does not have the needed accuracy. Define

$$h_n(t) = \min_A \log \sum_{x,s} e^{ts Q_A(x)}.$$

Proposition 5 (Exact annealed block estimate). *For all n, k and $t > 0$,*

$$h_{n+k}(t) \leq h_n(t) + h_k(t) - \log 2 + nk \log \cosh t.$$

Proof. For a within-block matrix A , write $Z_A^s = \sum_x e^{tsQ_A(x)}$. Replacing all coefficients of the second block by their negatives swaps its two one-sided partition sums. Of the two possible alignments, one satisfies

$$Z_A^+ Z_B^+ + Z_A^- Z_B^- \leq \frac{1}{2}(Z_A^+ + Z_A^-)(Z_B^+ + Z_B^-).$$

Choose every cross coefficient independently and uniformly. For fixed x, y, s , averaging the cross exponential gives $(\cosh t)^{nk}$. Therefore the expected augmented partition sum is at most the right side above times this factor, and some deterministic cross signing is no larger. Take logarithms and infima. $\square$

At $t = \beta/\sqrt{n+k}$ and $n/(n+k) \rightarrow \alpha$, the normalized cross cost tends to

$$\frac{\beta}{2}\alpha(1-\alpha),$$

which is nonzero for fixed β and grows in the zero-temperature limit. The direct annealed cross block is therefore a leading term, not an error.

Finally, the absolute value cannot be silently removed. If every edge from vertex 1 is positive and every other edge negative, fix $x_1 = 1$ and put $T = \sum_{i>1} x_i$. Then

$$Q_A(x) = \frac{n}{2} - \frac{(T-1)^2}{2}.$$

Hence $\max_x Q_A(x) = \lfloor n/2 \rfloor$, whereas $\max_x |Q_A(x)| = \binom{n}{2}$.

5 Covariance relaxations: two exact asymptotic limits

Let

$$\mathcal{E}_n = \{R \succeq 0 : \text{diag } R = \mathbf{1}\}$$

be the elliptope. The full arcsine expression is tautologically exact:

$$M(A) = \max_{R \in \mathcal{E}_n} \left| \frac{2}{\pi} \sum_{i < j} a_{ij} \arcsin R_{ij} \right|. \quad (8)$$

Indeed, Gaussian sign rounding makes every value on the right at most $M(A)$, while $R = xx^\top$ attains $Q_A(x)$. The nonlinearity therefore preserves the original problem rather than relaxing it.

Linearizing gives a model with a complete thermodynamic limit. Put

$$S(A) = \max_{R \in \mathcal{E}_n} \left| \frac{1}{2} \text{tr}(AR) \right|, \quad S_n = \min_A S(A).$$

Theorem 6 (Adversarial elliptope limit). *For every $n \geq 2$,*

$$S_n \geq \frac{1}{2}n\sqrt{n-1},$$

and

$$\boxed{\lim_{n \rightarrow \infty} \frac{S_n}{n^{3/2}} = \frac{1}{2}}.$$

At each symmetric conference order, equality holds in the finite bound.

Proof. Fix any sign matrix A , set $q = n - 1$, and define

$$R^\pm = \frac{1}{2} \left(I \pm \frac{A}{\sqrt{q}} \right)^2.$$

They are positive semidefinite. Since every row of A has squared norm q , their diagonals equal one. Moreover

$$R^+ - R^- = \frac{2A}{\sqrt{q}}.$$

The corresponding two linear SDP values differ by

$$\frac{1}{2} \operatorname{tr}(A(R^+ - R^-)) = n\sqrt{q}.$$

At least one has absolute value at least half this amount.

For the reverse asymptotic inequality, if $R \in \mathcal{E}_n$, then

$$|\operatorname{tr}(AR)| \leq \|A\|_{\text{op}} \operatorname{tr} R = n\|A\|_{\text{op}}.$$

A symmetric conference matrix has operator norm $\sqrt{n-1}$, so it attains equality. For general n , the prime number theorem in arithmetic progressions supplies a prime $q \equiv 1 \pmod{4}$ with $q+1 \geq n$ and $q+1 = n(1+o(1))$ [14]. Take a principal n -by- n submatrix of the symmetric Paley conference matrix of order $N = q+1$. Compression does not increase its operator norm $\sqrt{N-1}$, giving $S_n \leq n\sqrt{N-1}/2 = (1/2 + o(1))n^{3/2}$. $\square$

Rank-one $R = xx^\top$ show $S(A) \geq M(A)$. The missing transfer is an asymptotically vanishing integrality gap, which is not supplied by standard Grothendieck-type constant-factor rounding [5, 8]. The gap is already strict at order six: the exact computation below gives $F(6) = 5$, whereas Theorem 6 and a conference matrix give $S_6 = 3\sqrt{5}$.

There is also a sharp barrier for the linear Gaussian-sign method. Let $\Lambda(A)$ be the maximum of

$$\sum_{i < j} a_{ij}(R_{ij}^+ - R_{ij}^-)$$

over $R^\pm \in \mathcal{E}_n$, subject to the edgewise orientations $a_{ij}(R_{ij}^+ - R_{ij}^-) \geq 0$.

Theorem 7 (Oriented covariance barrier). *For every sign matrix A ,*

$$M(A) \geq \frac{\Lambda(A)}{\pi}, \quad \Lambda(A) \geq n\sqrt{n-1}.$$

Moreover

$$\lim_{n \rightarrow \infty} \frac{\min_A \Lambda(A)}{n^{3/2}} = 1.$$

Proof. Gaussian sign rounding of R^+ and R^- , followed by the arcsine law, gives two expected energies. Since $\arcsin'(r) \geq 1$ and each edge difference has the prescribed orientation, their difference is at least $2\Lambda(A)/\pi$. Both expectations lie in $[-M(A), M(A)]$, proving the first inequality.

The square covariances used in Theorem 6 satisfy the orientation constraints and have objective $n\sqrt{n-1}$. Conversely,

$$\Lambda(A) = \frac{1}{2} \operatorname{tr}(A(R^+ - R^-)) \leq n\|A\|_{\text{op}}.$$

Conference matrices attain equality. The same prime-number-theorem and Paley principal-submatrix construction used in Theorem 6 gives the matching asymptotic upper bound at every order. $\square$

Thus the optimized two-covariance, edgewise-oriented linear certificate Λ defined above has adversarial asymptotic lower-bound constant exactly $1/\pi$. Improving it requires information beyond this specific linearization; the theorem does not exclude more general covariance certificates.

6 The augmented cut code and its exact dual

Encode signs as bits via $(-1)^b$. Let $j \in \mathbb{F}_2^m$ be the all-one edge word and

$$(\delta u)_{ij} = u_i + u_j.$$

Then the augmented code in (2) is

$$D_n = \{\delta u + sj : u \in \mathbb{F}_2^n, s \in \mathbb{F}_2\}.$$

Theorem 8 (Linear code and dual). *For $n \geq 3$, D_n is a binary $[m, n]$ linear code and*

$$D_n^\perp = \{H \subseteq E(K_n) : \deg_H(v) \equiv 0 \pmod{2} \text{ for all } v, |H| \equiv 0 \pmod{2}\}.$$

For $n \geq 4$, the dual is spanned by 4-cycles, has dimension $m - n$, and minimum distance four.

Proof. The cut space has dimension $n - 1$. Its words have even parity on every triangle, whereas j has odd triangle parity, so adjoining j raises the dimension by one. A graph is orthogonal to every cut exactly when every vertex has even degree, and it is orthogonal to j exactly when its number of edges is even.

The cycle space is generated by triangles through one fixed vertex. An even-weight cycle-space word is a sum of an even number of these basis triangles. The sum of two sharing a nonfixed vertex is a 4-cycle. Two that share only the fixed vertex can be paired through an intermediate basis triangle and expressed as two 4-cycles. This proves generation. A nonzero even-cardinality Eulerian graph has at least four edges, and a 4-cycle exists. $\square$

Counting cuts by the size of one shore gives

$$W_{D_n}(z) = \frac{1}{2} \sum_{k=0}^n \binom{n}{k} \left(z^{k(n-k)} + z^{m-k(n-k)} \right). \quad (9)$$

Let $a_e = (-1)^{y_e}$ and define the exact augmented partition function

$$Z_y(\beta) = \sum_{d \in D_n} \exp(\beta[m - 2 \text{wt}(y + d)]).$$

Its ground state is $M(A)$.

Theorem 9 (Signed-Eulerian MacWilliams formula). *For $n \geq 3$,*

$$Z_y(\beta) = 2^n (\cosh \beta)^m P_y(\tanh \beta),$$

where

$$P_y(t) = \sum_{H \in D_n^\perp} (-1)^{|H \cap y|_t |H|}.$$

For $0 \leq t < 1$, $P_y(t) > 0$, and uniformly in y ,

$$M(A) \leq \frac{1}{\beta} \log Z_y(\beta) \leq M(A) + \frac{n \log 2}{\beta}.$$

Proof. The coset MacWilliams identity is

$$W_{y+D_n}(z) = 2^{n-m}(1+z)^m \sum_{H \in D_n^\perp} (-1)^{|H \cap y|} \left(\frac{1-z}{1+z} \right)^{|H|}.$$

Multiply by $e^{\beta m}$ and substitute $z = e^{-2\beta}$. This gives the formula because $e^\beta(1 + e^{-2\beta}) = 2 \cosh \beta$ and the final ratio is $\tanh \beta$. Positivity follows because the left side is a positive partition sum. The last display compares its largest summand with a sum of $|D_n| = 2^n$ summands. $\square$

For fixed $\beta > 0$, the minimized free energy in Theorem 9, divided by $n^{3/2}$, differs from $F(n)/n^{3/2}$ by $O(n^{-1/2})$. Thus the exact unresolved term is

$$\min_y \log P_y(\tanh \beta).$$

It must cancel the explicit $m \log \cosh \beta$ term to $n^{3/2}$ accuracy. Averaging $P_y(t)$ over all y gives one and yields only

$$F(n) \leq n \sqrt{(n-1) \log 2},$$

the usual sphere/union-bound constant, weaker than $1/2$.

Puncturing also stops at the old scale. For $n \geq 4$, deleting the final star maps D_n onto D_{n-1} , and the two lifts of each word have complementary star coordinates. Therefore

$$\rho(D_{n-1}) \leq \rho(D_n) \leq \rho(D_{n-1}) + \left\lfloor \frac{n-1}{2} \right\rfloor,$$

equivalently

$$F(n-1) + ((n-1) \bmod 2) \leq F(n) \leq F(n-1) + n-1.$$

The $n = 3$ instance of these inequalities is checked directly; the fiber there has four rather than two words. Across a two-block partition, the cross lifts are complementary rank-one sign matrices. On the dual side, the $\binom{n}{2} \binom{k}{2}$ alternating cross-block 4-cycles already contribute macroscopically at mean-field scaling. Product bounds that take absolute values discard the cancellation needed here.

7 High-temperature data do not truncate uniformly

Let X be uniform on $\{-1, 1\}^n$, let $Y = Q_A(X)$, and use the symmetrized partition function $\mathbb{E} \cosh(tY)$, equivalently the augmented state sY . Direct Walsh counting gives

$$\mathbb{E} Y^2 = m, \quad \mathbb{E} Y^4 = 3m^2 - 2m + 24 \sum_{C_4} \prod_{e \in C_4} a_e, \quad (10)$$

where the sum is over undirected 4-cycles. Indeed, a monomial in the fourth power survives exactly when every vertex has even incidence. Besides one edge four times and two edges twice, the only case is a simple 4-cycle, with multinomial factor 24.

Equivalently,

$$\kappa_4(Y) = 3 \operatorname{tr}(A^4) - 2n(n-1)(3n-4), \quad \operatorname{tr}(A^4) = n(n-1)^2 + \sum_{i \neq j} (A_{ij}^2)^2. \quad (11)$$

The second identity is $\operatorname{tr}(A^4) = \sum_{ij} (A_{ij}^2)^2$; the first follows by separating the closed walks with four distinct vertices in (10).

The expansion

$$\log \mathbb{E} \cosh(tY) = \frac{mt^2}{2} + \frac{\kappa_4(Y)t^4}{24} + O(t^6)$$

shows that, for each fixed n , all sufficiently high-temperature minimizers of this symmetrized partition function minimize $\text{tr}(A^4)$. The symmetrization matters: the one-sided partition can have a signing-dependent cubic triangle term. Spectral Cauchy–Schwarz gives

$$\text{tr}(A^4) \geq \frac{\text{tr}(A^2)^2}{n} = n(n-1)^2.$$

For a Paley principal submatrix of order n inside a conference matrix of order $N = n(1 + o(1))$,

$$\text{tr}(A^4) \leq \|A\|_{\text{op}}^2 \text{tr}(A^2) \leq (N-1)n(n-1).$$

This gives the reverse asymptotic inequality, so

$$\lim_n \frac{\min_A \text{tr}(A^4)}{n^3} = 1, \quad \lim_n \frac{\min_A \kappa_4(Y)}{n^3} = -3.$$

This fourth-order limit does not determine the next term. Exhausting all root-normalized order-seven signings gives

$$\min_A \text{tr}(A^4) = 342,$$

while the trace-minimizing signings have three distinct exact sixth moments,

$$50,781, \quad 53,661, \quad 59,421.$$

Thus fourth-order data alone are insufficient. No uniform remainder theorem connecting this high-temperature expansion to the required ground-state regime was obtained.

8 Ordinary graphons and empirical spectra lose the functional

For a signed matrix A , let W_A be its step graphon and normalize

$$\|W_A\|_{\square} = \frac{1}{n^2} \max_{S, T \subseteq [n]} \left| \sum_{i \in S, j \in T} a_{ij} \right|.$$

Theorem 10 (Cut-limit obstruction). *If $M(A_n) = O(n^{3/2})$, then $W_{A_n} \rightarrow 0$ in cut norm. There are two deterministic sequences satisfying this hypothesis and having the same zero graphon limit, while their normalized Boolean maxima are separated by at least $3/2$.*

Proof. For $x, y \in \{-1, 1\}^n$, split the coordinates according to whether $x_i = y_i$. Writing $z^+, z^- \in \{-1, 0, 1\}^n$ for the corresponding restrictions of y , symmetry of A cancels cross terms and gives

$$x^T A y = 2Q_A(z^+) - 2Q_A(z^-).$$

A multilinear quadratic form attains its maximum absolute value on $[-1, 1]^n$ at a Boolean vertex, so every sign bilinear form is bounded by $4M(A)$. Expressing two indicator vectors as affine combinations of sign vectors then gives

$$\|W_A\|_{\square} \leq \frac{4M(A)}{n^2}.$$

Take an infinite sequence of Paley conference matrices C_N . Their normalized Boolean maxima have limsup at most $1/2$, and their graphons tend to zero. Choose $k = \lfloor 2N^{3/4} \rfloor$ vertices and change every coefficient inside that set to $+1$, obtaining B_N . At most $\binom{k}{2} = O(N^{3/2})$ edges change, so $M(B_N) = O(N^{3/2})$ and the graphon perturbation tends to zero in L^1 . Fix the planted spins to $+1$ and average all outside spins independently. Every cross and outside term has mean zero, while the planted block contributes $\binom{k}{2}$. Some outside assignment therefore has at least that energy, and

$$\liminf_N \frac{M(B_N)}{N^{3/2}} \geq 2.$$

The separation from the conference limsup is at least $3/2$. $\square$

The usual dense graph topology therefore collapses every target-scale near-optimizer to the same point. This is not repaired by keeping only a limiting empirical spectrum. At dense scaling, every empirical law of A/n tends to δ_0 , since its average squared eigenvalue is $(n-1)/n^2$. At fluctuation scaling there is a stronger obstruction.

Proposition 11 (Empirical-spectral obstruction). *For every fixed $d > 1/2$, there are conference matrices C_N and sign matrices B_N such that*

$$W_2(\mu_{B_N/\sqrt{N}}, \mu_{C_N/\sqrt{N}}) \longrightarrow 0,$$

but

$$\liminf_N \frac{M(B_N)}{N^{3/2}} \geq d, \quad \limsup_N \frac{M(C_N)}{N^{3/2}} \leq \frac{1}{2}.$$

Proof. Let $q_N = Q_{C_N}(\mathbf{1})$. The spectral bound gives $|q_N| \leq N\sqrt{N-1}/2$. Flip negative coefficients to positive until the all-one energy is at least $dN^{3/2}$. This needs $O(N^{3/2})$ edges, and there are $\Theta(N^2)$ available. Therefore $\|B_N - C_N\|_F^2 = O(N^{3/2})$. Hoffman–Wielandt implies

$$W_2^2(\mu_{B_N/\sqrt{N}}, \mu_{C_N/\sqrt{N}}) \leq \frac{\|B_N - C_N\|_F^2}{N^2} = O(N^{-1/2}).$$

The energy inequalities follow from the construction and the conference spectral bound. $\square$

This excludes dense graphons and limiting empirical spectral distributions, not spectral edges, eigenvectors, or a richer Gaussian-process limit.

For comparison, iid signings are rigorously worse than the conference upper constant.

Proposition 12 (Greedy energy of an iid signing). *On one infinite iid Rademacher edge array A_n , almost surely*

$$\liminf_n \frac{M(A_n)}{n^{3/2}} \geq \frac{2}{3} \sqrt{\frac{2}{\pi}} = 0.531923 \dots > \frac{1}{2}.$$

Proof. Choose spins sequentially by

$$x_1 = 1, \quad x_j = \operatorname{sgn} \sum_{i < j} a_{ij} x_i.$$

The resulting energy is

$$Q_{A_n}(x) = \sum_{j=2}^n \left| \sum_{i < j} a_{ij} x_i \right| = \sum_{k=1}^{n-1} Y_k.$$

Before column j is exposed, the earlier spins are measurable with respect to earlier columns. Multiplying the fresh iid column signs by those predictable spins preserves their iid Rademacher law. Hence the Y_k are independent and Y_k has the law of $|\varepsilon_1 + \dots + \varepsilon_k|$.

The central limit theorem and uniform integrability give $\mathbb{E}Y_k/\sqrt{k} \rightarrow \sqrt{2/\pi}$. A Riemann-sum argument therefore yields

$$\frac{1}{n^{3/2}} \sum_{k < n} \mathbb{E}Y_k \longrightarrow \sqrt{\frac{2}{\pi}} \int_0^1 \sqrt{t} dt = \frac{2}{3} \sqrt{\frac{2}{\pi}}.$$

Changing one input sign changes Y_k by at most two, so its centered moment generating function is bounded by $\exp(\lambda^2 k/2)$. Independence gives an exponential tail $2e^{-c_\varepsilon n}$ for deviations of $\sum_{k < n} Y_k$ by $\varepsilon n^{3/2}$. This is summable; Borel–Cantelli completes the proof. $\square$

This is not a lower bound on $F(n)$; it shows that typical iid signings are not asymptotically optimal.

9 Exact computation through order fourteen

Switching makes every edge incident with vertex zero positive. The remaining negative coefficients form a graph on $n - 1$ vertices, and vertex permutations reduce the search to one unlabeled graph in each isomorphism class. For a residual negative-edge graph G and a negative-spin set S not containing the root, the exact energy is

$$\binom{n}{2} - 2|E(G)| - 2|S|(n - |S|) + 4|\delta_G(S)|.$$

Exhausting nauty’s `geng` streams gives Table 1.

Table 1: Exact computational values.

n	2	3	4	5	6	7	8	9	10	11	12	13	14
$F(n)$	1	3	4	4	5	9	10	12	13	17	18	20	21
$F(n)/n^{3/2}$	.3536	.5774	.5000	.3578	.3402	.4860	.4419	.4444	.4111	.4660	.4330	.4267	.4009

Completeness is checked against the unlabeled-graph counts and hashes of the full graph streams. The reduced formula is checked by direct evaluation; all labeled signings are independently enumerated through $n = 6$; optional NetworkX and Z3 routes use different decoders and optimization mechanisms. Z3 independently gives UNSAT/SAT pairs certifying the displayed values for $n = 7, 8, 9$. At $n = 10$, the lower certificate is the exhaustive nauty search. At order 11, the full stream of 12,005,168 unlabeled residual graphs is hashed and exactly filtered to the classes that can have maximum at most 15. A reduced stream with $20 \leq |E(G)| \leq 22$ and maximum degree at most six contains 2,153,606 records and reproduces the zero-survivor result with the same evaluator. The two stream hashes and the exact cut computation are recorded in `verification/research_order11_certify.py`. Deterministic lower-scan samples at edge counts 20, 21, and 22 are also recomputed by an adjacency formula. The order-11 graph6 witness `ICRbcbzQM0` has maximum 17. Puncturing and energy parity then give $F(12) \geq 18$, while the witness `JWUuDOR\K{?}` has maximum 18. Completeness at orders 10 and 11 trusts nauty; no solver timeout is used in either result.

At order 13, a separate eight-shard certificate hashes and scans all 1,018,997,864 unlabeled residual graphs on eleven vertices. The fixed-order scanner completely enumerates the 2^{11} projective

spin classes unless it has already witnessed $|Q| \geq 19$. Since order-12 energies are even, precisely the records with maximum at most 18 survive. There are exactly two rooted survivors, JCpVdXyxpz? and JCpdUg{[dM?. Independent full-cube evaluation gives maximum 18 for both, and direct enumeration of all projective incident columns gives one-vertex extension minimum 24 for both. Every other order-12 predecessor has maximum at least 20, so the Bellman extension identity gives $F(13) \geq 20$. Every one-point deletion of the Paley conference matrix C_{14} has maximum 20, proving

$$F(13) = 20.$$

Heredity gives $F(14) \geq 20$, and every order-14 energy is odd. Direct evaluation gives $M(C_{14}) = 21$, hence

$$F(14) = 21.$$

The order-13 certificate checks producer and scanner exit statuses and pins the byte count, record count, and SHA-256 digest of every nauty shard. Its explicit remaining trust boundary is nauty's completeness for unlabeled graphs; it does not use solver statuses or stored scan logs.

The Paley conference matrix of order ten over $\mathbb{F}_9$ has exact Boolean maximum 15, strictly worse than $F(10) = 13$. The two optimal switching classes at order ten have different characteristic polynomials. Thus neither conference structure nor a unique spectrum describes all finite optimizers. These finite facts do not imply a limiting constant.

10 Finite non-heredity and an exact weighted covering state

For fixed signings A and B , define

$$J(A, B) = \min_{C \in \{-1, 1\}^{s \times k}} M \begin{pmatrix} A & C \\ C^\top & B \end{pmatrix}, \quad E(B) = J([0], B).$$

Complete cross-block enumeration over the optimal switching-permutation classes through total order ten gives, in particular,

$$\min_{M(A)=F(2), M(B)=F(8)} J(A, B) = 15 > F(10) = 13. \quad (12)$$

Hence no optimal order-10 signing contains an optimal order-8 principal submatrix. The smallest possible internal sacrifice repairs the obstruction: among the 1,044 root-normalized order-8 representatives, internal maximum 12 is necessary and sufficient for a $2 + 8$ completion of maximum 13. The same enumeration gives one-vertex extension distributions

n	distribution of $E(B)$ over optimal classes
7	$10^{\times 4}, 12^{\times 2}$
8	$12^{\times 2}$
9	$13^{\times 4}, 15^{\times 11}$
10	$17^{\times 1}, 19^{\times 1}$.

Thus the scalar $M(B)$ does not determine future composition cost.

There is, however, an exact coding-theoretic state for one-vertex extension. Let

$$\mathcal{P}_n = \{-1, 1\}^n / \{x \sim -x\}, \quad d_\pm([u], [v]) = \min\{d_H(u, v), d_H(u, -v)\}.$$

For an order- n signing B , put $M = M(B)$ and

$$w_B([x]) = \frac{M - |Q_B(x)|}{2}, \quad \rho_w(B) = \max_{[b] \in \mathcal{P}_n} \min_{[x] \in \mathcal{P}_n} (d_\pm([b], [x]) + w_B([x])).$$

The weights are nonnegative integers because every quadratic energy has the parity of $\binom{n}{2}$.

Theorem 13 (Weighted covering-radius Bellman identity). *For every order- n signing B ,*

$$E(B) = M(B) + n - 2\rho_w(B).$$

Equivalently, if

$$\delta_w(B) = \left\lfloor \frac{n}{2} \right\rfloor - \rho_w(B),$$

then

$$F(n+1) = \min_B (M(B) + (n \bmod 2) + 2\delta_w(B)),$$

where the minimum is over all order- n signings.

Proof. Let $b \in \{-1, 1\}^n$ be the incident signs and let t be the new spin. Pairing $t = 1$ and $t = -1$ gives, for every old spin vector x ,

$$\max_{t \in \{-1, 1\}} |Q_B(x) + t b \cdot x| = |Q_B(x)| + |b \cdot x|.$$

Since $|b \cdot x| = n - 2d_\pm([b], [x])$, the maximum over x for fixed b is

$$M + n - 2 \min_{[x] \in \mathcal{P}_n} (w_B([x]) + d_\pm([b], [x])).$$

Minimizing over $[b]$ proves the first identity. The equality $n - 2\lfloor n/2 \rfloor = n \bmod 2$ gives the deficit form. Finally, minimizing over all B and b enumerates every order- $(n+1)$ signing. $\square$

For every center $[b]$, some exact maximizer has projective distance at most $\lfloor n/2 \rfloor$. A configuration of weight greater than this can never attain the inner minimum. Consequently $E(B)$ depends only on the exact near-ground window

$$|Q_B(x)| \geq M(B) - 2 \left\lfloor \frac{n}{2} \right\rfloor.$$

The Bellman state has energy width at most n ; deeper configurations are rigorously irrelevant to one-vertex extension.

Let $X_B = \{[x] : |Q_B(x)| = M(B)\}$ and let $\rho_{\text{ext}}(B)$ be its ordinary covering radius in $\mathcal{P}_n$. The same proof, restricted to the maximizing configurations, gives the exact lower obstruction

$$E(B) \geq M(B) + n - 2\rho_{\text{ext}}(B), \quad \rho_w(B) \leq \rho_{\text{ext}}(B).$$

The exhaustive verifier independently recomputes both radii and the full extension values. At order nine, the four extendible classes have $(\rho_{\text{ext}}, \rho_w) = (4, 4)$, while the eleven failing classes have $(3, 3)$. The order-10 classes have radii $(3, 3)$ and $(2, 2)$, exactly explaining extension values 17 and 19. Exact maximizers are not a complete state: all six order-7 classes have $\rho_{\text{ext}} = 3$, but the $|Q_B| = 7$ layer lowers ρ_w to 2 in precisely the two classes with $E(B) = 12$. This is a finite counterexample to any cavity rule retaining only the ground-state code. The full energy-weighted projective landscape is exact for one step.

The scalar partition function is also insufficient. The two optimal order-nine graph6 records `G?qmaw` and `GCpbaw` have the same complete projective absolute-energy histogram

$$\#\{|Q_B| = 0, 4, 8, 12\} = (60, 111, 60, 25),$$

and therefore have identical values of $\sum_x e^{t|Q_B(x)|}$ and $\sum_{x,s} e^{tsQ_B(x)}$ for every real t . Nevertheless their extremizer covering radii are 4 and 3, and direct enumeration gives extension values 13 and 15.

Thus even the entire scalar free-energy curve loses the geometry required by a Bellman step. This does not refute a nonlocal minimax interpolation with a richer order parameter.

The verifier also exhausts all root-normalized residual graphs through order eight. If $\mathcal{B}_n$ is the coordinatewise Pareto-minimal set of pairs $(M(B), \delta_w(B))$, then

n	$\mathcal{B}_n$
2	$\{(1, 1)\}$
3	$\{(3, 0)\}$
4	$\{(4, 0)\}$
5	$\{(4, 0)\}$
6	$\{(5, 2), (7, 1), (9, 0)\}$
7	$\{(9, 0)\}$
8	$\{(10, 1), (12, 0)\}$.

Every point at order six gives $F(7) = 9$, and both points at order eight give $F(9) = 12$. Thus two units of additional internal energy can be exactly offset by one unit of covering-deficit improvement. This is finite evidence for a near-optimal Pareto family, not an asymptotic conclusion.

10.1 The nonlinear square-covariance gain

The earlier Gaussian proof used only the inequality $\arcsin'(u) \geq 1$. Retaining the discarded nonlinear term gives a stability theorem.

Theorem 14 (Nonlinear Gaussian-sign bound). *For $s \geq 0$, put*

$$D_s = 1 + s^2(n-1), \quad d_s = \frac{2s}{D_s}, \quad z_{ij,s} = \frac{s^2|(A^2)_{ij}|}{D_s}.$$

Then every order- n signing satisfies

$$M(A) \geq \frac{1}{\pi} \sum_{i < j} [\arcsin(z_{ij,s} + d_s) - \arcsin(z_{ij,s} - d_s)]. \quad (25)$$

Consequently

$$M(A) \geq \frac{n(n-1)}{\pi} \arcsin \frac{1}{\sqrt{n-1}}, \quad (26)$$

and, for $n \geq 3$,

$$M(A) \geq \frac{n(n-1)}{\pi} \arcsin \frac{1}{\sqrt{n-1}} + \frac{\text{tr}(A^4) - n(n-1)^2}{8\pi(n-1)(n-2)^{3/2}}. \quad (27)$$

Proof. The two matrices

$$R_s^\pm = \frac{(I \pm sA)^2}{D_s}$$

are correlation matrices. With $u_{ij} = s^2 a_{ij}(A^2)_{ij}/D_s$, the corresponding Gaussian sign roundings and the arcsine law give

$$\mathbb{E}Q_A(X^+) - \mathbb{E}Q_A(X^-) = \frac{2}{\pi} \sum_{i < j} [\arcsin(u_{ij} + d_s) - \arcsin(u_{ij} - d_s)].$$

The bracket is even in u_{ij} , and each expectation belongs to $[-M(A), M(A)]$, proving (25).

For $f_d(z) = \arcsin(z+d) - \arcsin(z-d)$ and $z \geq 0$,

$$f_d''(z) = h(z+d) - h(z-d), \quad h(t) = \frac{t}{(1-t^2)^{3/2}}.$$

Thus f_d is even and convex. At $s = 1/\sqrt{n-1}$ one has $D_s = 2$ and $d_s = 1/\sqrt{n-1}$, so $f_d(z) \geq f_d(0) = 2 \arcsin d$, which proves (26).

Write $q = n-1$. Since

$$h'(t) = \frac{1+2t^2}{(1-t^2)^{5/2}}$$

is even and increasing in $|t|$, $f_d''(z)$ is increasing for $z \geq 0$. For $d = 1/\sqrt{q}$,

$$f_d(z) \geq 2 \arcsin d + \frac{q}{(q-1)^{3/2}} z^2.$$

Finally $z_{ij} = |(A^2)_{ij}|/(2q)$ and

$$\sum_{i < j} z_{ij}^2 = \frac{\text{tr}(A^4) - nq^2}{8q^2},$$

which proves (27). □

The excess is exactly

$$\text{tr}(A^4) - n(n-1)^2 = \|A^2 - (n-1)I\|_F^2.$$

Hence a sequence attaining the asymptotic lower constant $1/\pi$ must obey $\|A_n^2 - (n-1)I\|_F = o(n^2)$. This is vanishing normalized Gram defect, not a conference-rigidity conclusion: a uniform random signing already satisfies this normalization in probability. The universal improvement in (26) is only $\Theta(\sqrt{n})$, so it does not change the leading constant. It does imply the new exact consequence

$$\boxed{F(21) \geq 32}.$$

Indeed (26) is greater than 30 and all order-21 energies are even. For an exact strictness certificate, use $\pi/14 < 11/49$ and $\sin t < t - t^3/6 + t^5/120$; rational squaring shows that the latter polynomial at $11/49$ has square less than $1/20$. Thus $\arcsin(1/\sqrt{20}) > \pi/14$.

10.2 The multivertex Bellman state and its entropy bound

For internal signings B, D of orders n, k , define

$$J(B, D) = \min_C M \begin{pmatrix} B & C \\ C^\top & D \end{pmatrix}, \quad L(B, D) = \max_{x, y} |Q_B(x) + Q_D(y)|.$$

In the projective Hamming cube $\mathcal{P}_{nk} = \{\pm 1\}^{n \times k} / \{C \sim -C\}$, let

$$\mathcal{R}_{n,k} = \{[xy^\top] : x \in \{-1, 1\}^n, y \in \{-1, 1\}^k\}$$

and give $R = [xy^\top]$ the integer weight

$$w_{B,D}(R) = \frac{L(B, D) - |Q_B(x) + Q_D(y)|}{2}.$$

Put

$$\rho_w^\square(B, D) = \max_{[C] \in \mathcal{P}_{nk}} \min_{R \in \mathcal{R}_{n,k}} [d_\pm([C], R) + w_{B,D}(R)].$$

Theorem 15 (Multivertex weighted Bellman identity). *One has*

$$J(B, D) = L(B, D) + nk - 2\rho_w^\square(B, D). \quad (28)$$

Thus, with

$$\delta_w^\square = \left\lfloor \frac{nk}{2} \right\rfloor - \rho_w^\square, \quad (29)$$

$$F(n+k) = \min_{B,D} [L(B, D) + (nk \bmod 2) + 2\delta_w^\square(B, D)].$$

Proof. Pairing y with $-y$ gives

$$\max_{\eta=\pm 1} |Q_B(x) + Q_D(y) + \eta x^\top C y| = |Q_B(x) + Q_D(y)| + |x^\top C y|.$$

Moreover $|x^\top C y| = nk - 2d_\pm([C], [xy^\top])$. Substitute $|Q_B + Q_D| = L - 2w$, maximize over the rank-one code, and minimize over $[C]$ to obtain (28). A weight-zero rank-one word is within projective distance $\lfloor nk/2 \rfloor$ of every center, so the deficit is nonnegative. Minimizing over the two internal blocks proves (29). $\square$

Only the internal window

$$|Q_B(x) + Q_D(y)| \geq L(B, D) - 2 \left\lfloor \frac{nk}{2} \right\rfloor$$

can affect this radius. Define its soft effective size by

$$\Xi(B, D) = \log \sum_{R \in \mathcal{R}_{n,k}} \exp \left(-\frac{2w_{B,D}(R)^2}{nk} \right).$$

Proposition 16 (Weighted entropy upper bound).

$$J(B, D) \leq L(B, D) + \sqrt{2nk(\Xi(B, D) + \log 4)}. \quad (30)$$

Proof. Choose C uniformly at random. Hoeffding's inequality and a union bound give

$$\Pr\{\exists R : |\langle C, R \rangle| > T + 2w(R)\} \leq 2 \exp \left(-\frac{T^2}{2nk} + \Xi(B, D) \right).$$

For $T = \sqrt{2nk(\Xi + \log 4)}$ this is at most $1/2$. Hence some C satisfies every displayed inequality, and $L - 2w(R) + |\langle C, R \rangle| \leq L + T$ for every R . Theorem 15 finishes the proof. $\square$

The generic bound $\Xi \leq (n+k-2) \log 2$ reproduces the known leading $\sqrt{nk(n+k)}$ block cost. The initially proposed replacement was the following constant-matching profile law:

$$L(B, D) + \sqrt{2nk(\Xi(B, D) + \log 4)} \leq (F(n)^{2/3} + F(k)^{2/3})^{3/2} + O((n+k)^{3/2-\varepsilon}). \quad (31)$$

This would imply power-saving near-subadditivity of $F^{2/3}$, which the standard blocking argument turns into convergence. The next proposition shows that this initially proposed target is actually impossible; it is retained here as an audited failed route.

There is also density-one control of the exact one-vertex state. Choose B_n attaining Theorem 13 and put $\sigma_n = M(B_n) - F(n)$ and $\delta_n = \delta_w(B_n)$. Exactly

$$F(n+1) - F(n) = \sigma_n + (n \bmod 2) + 2\delta_n.$$

Telescoping and $F(n) = O(n^{3/2})$ show

$$\sum_{n=N}^{2N} (\sigma_n + 2\delta_n) = O(N^{3/2}).$$

Thus, for every $g(N) \rightarrow \infty$, all but $O(N/g(N))$ indices in $[N, 2N]$ have a Bellman-optimal predecessor with $\sigma_n \leq g(N)\sqrt{N}$ and $\delta_n \leq \frac{1}{2}g(N)\sqrt{N}$. The scalar countermodel below shows that even replacing this density-one statement by uniform $O(\sqrt{n})$ control would not prove convergence.

10.3 Parity gain and the sharp limit of the Gaussian certificate

Put

$$\Delta(A) = \|A^2 - (n-1)I\|_F^2 = \sum_{i \neq j} (A^2)_{ij}^2.$$

Theorem 17 (Parity of the Gram defect). *Every order- n signing satisfies*

$$\Delta(A) \geq \begin{cases} n(n-1), & n \text{ odd}, \\ 2n(n-2), & n \equiv 0 \pmod{4}, \\ 0, & n \equiv 2 \pmod{4}. \end{cases}$$

These bounds are attained on infinite Paley-derived subsequences. In particular,

$$\boxed{F(20) \geq 30}.$$

Proof. For odd n , every off-diagonal entry of A^2 is a sum of the odd number $n-2$ of signs and therefore has square at least one.

Now let $n \equiv 0 \pmod{4}$, write $a_{ij} = (-1)^{e_{ij}}$, and put $d_i = \sum_{j \neq i} e_{ij} \pmod{2}$. Reduction modulo four gives

$$\frac{(A^2)_{ij}}{2} \equiv 1 + d_i + d_j \pmod{2}.$$

Thus $|(A^2)_{ij}| \geq 2$ for every pair with $d_i = d_j$. If r of the d_i are odd, the handshake lemma makes r even and

$$\Delta(A) \geq 8 \left[\binom{r}{2} + \binom{n-r}{2} \right] \geq 2n(n-2).$$

Conference matrices show that the third bound can vanish. Deleting one vertex from a conference matrix makes all off-diagonal Gram entries signs; deleting two makes exactly $n(n-2)/2$ ordered entries have magnitude two. This proves sharpness on the corresponding Paley orders.

For $n = 20$, Theorem 14 and $\Delta \geq 720$ give

$$M(A) \geq \frac{380}{\pi} \arcsin \frac{1}{\sqrt{19}} + \frac{90}{19\pi 18^{3/2}} > 28.$$

For exact strictness, take $x = 39/170 < 1/\sqrt{19}$, use $1/\sqrt{18} > 70/297$, $\arcsin x > x + x^3/6$, and $\pi < 22/7$. The remaining rational margin is

$$380 \left(x + \frac{x^3}{6} \right) + \frac{5}{19} \frac{70}{297} - 88 = \frac{8798941}{2772405900} > 0.$$

Order-20 energies are even, so $M(A) \geq 30$. □

The full parameter family admits an exact Jensen compression. With $q = n - 1$, $\varepsilon = \Delta(A)/(nq^3)$, and $0 \leq w < 1$, put

$$a = \frac{1+w}{2}\sqrt{\varepsilon}, \quad d = \frac{\sqrt{1-w^2}}{\sqrt{q}}.$$

Then

$$M(A) \geq \frac{nq}{2\pi} [\arcsin(a+d) - \arcsin(a-d)].$$

Indeed, after writing $w = (qs^2 - 1)/(qs^2 + 1)$, the edge offsets in (25) have mean square a^2 . For fixed d , the function

$$u \mapsto \arcsin(\sqrt{u} + d) - \arcsin(\sqrt{u} - d)$$

is convex. To see this, if $f_d(z)$ denotes the arcsine difference, then

$$\frac{d^2}{du^2} f_d(\sqrt{u}) = \frac{zf_d''(z) - f_d'(z)}{4z^3}.$$

The numerator vanishes at zero and has derivative $zf_d'''(z) \geq 0$, because $h'(v)$ is even and increasing in $|v|$ for $h(v) = v(1 - v^2)^{-3/2}$. The number a is the root-mean-square of the edge offsets, so it remains in the correlation domain. Jensen proves the claim; endpoints follow by continuity. Consequently, if $\varepsilon(A_n) \rightarrow \varepsilon < 1$,

$$\liminf_n \frac{M(A_n)}{n^{3/2}} \geq \frac{1}{\pi} \sqrt{\frac{2}{1 + \sqrt{1 - \varepsilon}}}.$$

For fixed w , the limiting factor before optimization is

$$\frac{1}{\pi} \sqrt{\frac{1 - w^2}{1 - \varepsilon(1 + w)^2/4}}.$$

Here the square root multiplies $1/\pi$; optimizing its square gives $1 + w = 2/(1 + \sqrt{1 - \varepsilon})$ and the preceding formula.

Proposition 18 (Sharp barrier for the square-covariance family). *If $B_s(A)$ is the right side of (25) and $G_n = \min_A \sup_{s \geq 0} B_s(A)$, then*

$$\lim_{n \rightarrow \infty} \frac{G_n}{n^{3/2}} = \frac{1}{\pi}.$$

Proof. Taking $s = 1/\sqrt{n-1}$ proves the lower bound. For the reverse inequality, take an order- n principal block of a Paley conference matrix of order $n + r$, where $r = o(n)$. If U is the deleted block, then $|(A^2)_{ij}| = |(UU^\top)_{ij}| \leq r$. Uniformly in s ,

$$z_{ij,s} \leq \frac{r}{n-1} = o(1), \quad d_s \leq \frac{1}{\sqrt{n-1}}.$$

Writing each arcsine difference as an integral bounds it by $2d_s(1 + o(1))$, uniformly in s . Summation gives $\sup_s B_s(A) \leq (1/\pi + o(1))n^{3/2}$. $\square$

10.4 Failure of the weighted union bound and an exact orbit state

Proposition 19 (Universal entropy floor). *For every pair B, D ,*

$$L + \sqrt{2nk(\Xi + \log 4)} \geq \sqrt{2nk(n+k) \log 2}.$$

Proof. Since $|\mathcal{R}_{n,k}| = 2^{n+k-2}$ and $0 \leq w(R) \leq L/2$,

$$\Xi \geq (n+k-2) \log 2 - \frac{L^2}{2nk}.$$

Put $b = \sqrt{2nk(n+k) \log 2}$. If $L \leq b$, the left side is at least $L + \sqrt{b^2 - L^2} \geq b$; if $L > b$, the result is immediate. $\square$

For balanced blocks the floor is $2\sqrt{\log 2} N^{3/2}$, whereas the desired right side in (31) is at most $(\sqrt{2} + o(1))N^{3/2}$. Thus (31) fails universally by the leading constant $2\sqrt{\log 2} - \sqrt{2} = 0.250895\dots$. Even retaining the discarded cross term in the same Hoeffding sum cannot help: for $K \geq L$,

$$2 \sum_R \exp\left(-\frac{(K - h(R))^2}{2nk}\right) \geq 2^{n+k-1} \exp\left(-\frac{K^2}{2nk}\right).$$

There is an exact dependence-sensitive replacement. Regard $G = \mathcal{R}_{n,k}$ as a group under entrywise multiplication. For a cross seed C_0 put

$$c(U) = |\langle C_0, U \rangle|, \quad h(R) = |Q_B(x) + Q_D(y)|.$$

The row-and-column switching orbit $C_g = C_0 \odot g$ satisfies

$$J_{\text{orb}}(C_0; B, D) = \min_{g \in G} \max_{R \in G} [h(R) + c(gR)].$$

For $K \geq L$, let $C_t = \{U : c(U) > t\}$. Its bad shifts are exactly

$$\mathcal{B}_K = \bigcup_{R \in G} R C_{K-h(R)}.$$

Hence the orbit succeeds if and only if $\mathcal{B}_K \neq G$. This weighted sumset noncoverage problem on $\mathbb{F}_2^{n+k-2}$ retains the intersections lost by the union bound.

10.5 The exact scalar Bellman wall

Let

$$r_n = F(n+1) - F(n) - (n \bmod 2) = \sigma_n + 2\delta_n.$$

The normalized dyadic cost

$$A_N = N^{-3/2} \sum_{n=N}^{2N-1} r_n$$

is an exact diagnostic:

$$\frac{F(n)}{n^{3/2}} \text{ converges} \iff A_N \text{ converges}.$$

If $A_N \rightarrow \Lambda$, the limit is $\Lambda/(2^{3/2} - 1)$. Indeed, with $c_N = F(N)/N^{3/2}$ and $q = 2^{-3/2}$,

$$c_{2N} = qc_N + qA_N + O(N^{-1/2}).$$

The forward implication is immediate. Conversely, for arbitrary n and $m = \lfloor n/2 \rfloor$, the same relation and $F(2m+1) - F(2m) = O(m)$ give $c_n = q(c_m + \Lambda) + o(1)$. If $C = \Lambda/(2^{3/2} - 1)$ and $e_n = c_n - C$, boundedness yields $\limsup |e_n| \leq q \limsup |e_n|$, hence convergence at all orders.

Uniform magnitude control of r_n is not enough. Define

$$g(t) = t^{3/2} \left[\frac{2}{5} + \frac{1}{25} \sin \log \log(t + e^2) \right],$$

use

$$f(1), \dots, f(10) = 0, 1, 3, 4, 4, 5, 9, 10, 12, 13,$$

and for $n \geq 10$ let $f(n+1) - f(n)$ be the least positive integer congruent to n modulo two and at least $g(n+1) - g(n)$. Since

$$0.52\sqrt{t} \leq g'(t) \leq 0.68\sqrt{t} \quad (t \geq 10),$$

the resulting Bellman residual is nonnegative, even, and at most $3\sqrt{n}$. Moreover $f(n) = 13 + g(n) - g(10) + O(n)$, so its normalized liminf and limsup are 0.36 and 0.44: the phase $\log \log(n + e^2)$ is unbounded and its successive increments tend to zero. For $n \geq 10$,

$$f(n) \geq 13 + g(n) - g(10) \geq 0.36n^{3/2} + 13 - 0.44 \cdot 10^{3/2} > \frac{n^{3/2}}{\pi},$$

and the finite prefix is direct. The increment ceiling gives the trivial upper bound. Positive tail increments, together with the finite prefix, also give both scalar hereditary consequences proved earlier; the complete inequality check is in `RESEARCH.CONTINUATION.md`. Thus no theorem using only those scalar facts can force convergence. A sufficient replacement is

$$\frac{1}{N} \sum_{n \leq N} \frac{r_n}{\sqrt{n}} \rightarrow a,$$

because Abel summation then gives $F(N)/N^{3/2} \rightarrow 2a/3$.

10.6 Complete order-nine weighted geometry

Exhausting all 12,346 root-normalized order-nine signings gives the new full frontier

$$\mathcal{B}_9 = \{(12, 0)\}.$$

The 55 root records with $M = 12$ split as 20 with deficit zero and 35 with deficit one; their 15 switching-permutation classes split as four and eleven.

The records `GH0gmo` and `Gxd?Dc` give a stronger collision. Both have $M = 14$, projective energy histogram

$$\#\{|Q| = 2, 6, 10, 14\} = (124, 85, 37, 10),$$

and ordered pair-distance enumerator $(10, 16, 14, 20, 40)$ for their ten exact maximizers. Their profiles $(\rho_{\text{ext}}, \rho_{\text{w}}, E)$ are, however, $(4, 4, 15)$ and $(3, 3, 17)$. Hence the energy histogram plus the two-point overlap law of exact maximizers still does not determine extension.

Across the full rooted catalogue, ten histogram groups containing 874 records mix deficits. Adding the maximizer pair-distance law leaves three mixed groups containing 112 records. The complete energy-coloured two-point law

$$T_B(e, f, r) = \#\{([x], [y]) : |Q_B(x)| = e, |Q_B(y)| = f, d_{\pm}([x], [y]) = r\}$$

separates all deficits at order nine. This is a finite computation, not an asymptotic sufficiency claim. The verifier asserts the nauty stream count and digest and recomputes every extension both directly and from the weighted radius.

10.7 Complete order-ten finite-temperature phases

For order ten, let h_j count projective spins with $|Q_A| = 2j + 1$. Put

$$z = 4 \sinh^2 t, \quad R_0 = 1, \quad R_1 = z + 1, \quad R_{j+1} = (z + 2)R_j - R_{j-1}.$$

Then $\cosh((2j + 1)t) = \cosh(t)R_j(z)$, and the augmented partition function is $4 \cosh(t)P_H(z)$ with $P_H(z) = \sum_j h_j R_j(z)$.

The complete set of 274,668 root-normalized records has 6,012 distinct absolute-energy histograms. Every one of their polynomials coefficientwise dominates at least one of

$$\begin{aligned} P_0 &= 512 + 2816z + 4512z^2 + 3680z^3 + 2120z^4 + 792z^5 + 156z^6 + 12z^7, \\ P_1 &= 512 + 2816z + 5280z^2 + 4704z^3 + 2232z^4 + 576z^5 + 76z^6 + 4z^7, \\ P_2 &= 512 + 2816z + 5792z^2 + 5600z^3 + 2520z^4 + 520z^5 + 40z^6. \end{aligned}$$

Their respective nonzero energy histograms are

$$\begin{aligned} (h_0, h_3, h_4, h_7) &= (180, 180, 140, 12), \\ (h_0, \dots, h_7) &= (108, 88, 96, 84, 60, 48, 24, 4), \\ (h_1, h_2, h_5, h_6) &= (200, 192, 80, 40). \end{aligned}$$

The exact differences factor as

$$\begin{aligned} P_0 - P_1 &= 8z^2(z - 2)(z + 1)(z + 3)(z + 4)^2, \\ P_1 - P_2 &= 4z^2(z + 4)^2(z^3 + z^2 - 10z - 8). \end{aligned}$$

The cubic has one positive root $\zeta = 3.083872359\dots$. It follows that, for $t > 0$, the global minimizing histogram is P_0 until $t = \operatorname{arsinh}(1/\sqrt{2}) = 0.658478948\dots$, then P_1 until $t = \operatorname{arsinh}(\sqrt{\zeta}/2) = 0.792460762\dots$, and then P_2 . Adjacent phases tie at the thresholds; at $t = 0$ every signing ties. The maxima of the three phases are 15, 15, 13, respectively, and the first phase is a conference signing. Thus the adversarial optimizer genuinely changes with temperature. This is an exact finite catalogue result, not an asymptotic phase-transition theorem.

11 Negative replicas and exact parameter transport

For uniform disorder A of order n , define

$$f_{n,t}(A) = \frac{Z_A(t)}{2^{n+1}(\cosh t)^{m_n}}, \quad \mathcal{G}_n(q, t) = \log \mathbb{E}_A f_{n,t}(A)^{-q}.$$

The denominator is the annealed mean of $Z_A(t)$.

Theorem 20 (Negative-replica supermultiplicativity). *For every $n, k \geq 1$, $q > 0$, and $t \in \mathbb{R}$,*

$$\mathcal{G}_{n+k}(q, t) \geq \mathcal{G}_n(q, t) + \mathcal{G}_k(q, t). \quad (\text{NR})$$

Proof. Fix internal signings A, B and let C be a uniform $n \times k$ cross signing. Write

$$U_A^\sigma = \sum_x e^{t\sigma Q_A(x)}, \quad U_B^\sigma = \sum_y e^{t\sigma Q_B(y)}.$$

Independent averaging of the cross edges gives

$$\mathbb{E}_C Z_{A,B,C}(t) = (\cosh t)^{nk} P, \quad P = U_A^+ U_B^+ + U_A^- U_B^-.$$

Replacing B by $-B$ gives $P' = U_A^+ U_B^- + U_A^- U_B^+$, and $P + P' = Z_A Z_B$. Convexity of $z \mapsto z^{-q}$, first over C and then over the two orientations, yields

$$\begin{aligned} & \frac{1}{2} \left(\mathbb{E}_C Z_{A,B,C}^{-q} + \mathbb{E}_C Z_{A,-B,C}^{-q} \right) \\ & \geq \frac{(\cosh t)^{-qnk}}{2} (P^{-q} + P'^{-q}) \\ & \geq 2^q (\cosh t)^{-qnk} (Z_A Z_B)^{-q}. \end{aligned}$$

Uniform disorder is invariant under $B \mapsto -B$. After averaging A, B , the factors 2^q and $(\cosh t)^{-qnk}$ cancel exactly against the normalizers because $m_{n+k} = m_n + m_k + nk$. This proves (NR). At $t = 0$ there is equality. $\square$

Put $u = \tanh t$ and write $f_{n,u} = f_{n, \operatorname{arctanh} u}$ and $\mathcal{G}_n(q, u) = \mathcal{G}_n(q, \operatorname{arctanh} u)$. The signed even-Eulerian expansion above gives

$$f_{n,u} = T_{u/v} f_{n,v} \quad (0 < u < v < 1),$$

where T_ρ is the Boolean noise operator on disorder space. Sharp reverse hypercontractivity for positive functions gives, when $Q' > Q > 0$,

$$\|T_\rho f\|_{-Q'} \geq \|f\|_{-Q}, \quad \rho \leq \sqrt{\frac{1+Q}{1+Q'}} [10].$$

Since $\|f\|_{-Q} = \exp[-\mathcal{G}_n(Q, u)/Q]$ in this parametrization, on the exact invariant curve $(1+Q')u'^2 = (1+Q)u^2$ this says

$$\frac{\mathcal{G}_n(Q', u')}{Q'} \leq \frac{\mathcal{G}_n(Q, u)}{Q}. \quad (13)$$

This is precisely the curve produced by splitting an extensive system, but the inequality points in the wrong direction for closing (NR).

Fix $\beta, \theta > 0$ and, for sufficiently large r , set

$$t_r = \frac{\beta}{\sqrt{r}}, \quad q_r = \frac{\theta}{\tanh^2 t_r} - 1 = \frac{\theta}{\beta^2} r + \frac{2\theta}{3} - 1 + O(r^{-1}).$$

The first proposed repair was a power-saving upper bound on

$$\Delta_n(\beta, \theta) = \frac{q_{2n}}{q_n} \mathcal{G}_n(q_n, t_n) - \mathcal{G}_n(q_{2n}, t_{2n}). \quad (\text{PT-defect})$$

Such a bound would have produced balanced almost-superadditivity. It is false.

Theorem 21 (Leading scalar transport gap). *Put*

$$U = \limsup_{n \rightarrow \infty} \frac{F(n)}{n^{3/2}}.$$

Then

$$\liminf_{n \rightarrow \infty} \frac{\Delta_n(\beta, \theta)}{n^2} \geq \frac{2\theta}{\beta^2} \left[\frac{\beta^2}{8} - \beta \left(1 - \frac{1}{\sqrt{2}} \right) U - \log 2 \right] - \log 2. \quad (14)$$

In particular, the audited bound $U \leq 1/2$ gives

$$\liminf_n \frac{\Delta_n(4, 8)}{n^2} \geq \sqrt{2} - 2 \log 2 = 0.027919201253 \dots > 0.$$

Thus no power-saving estimate $\Delta_n = O(n^{2-\delta})$ can hold uniformly.

Proof. Write $m = m_n$ and $f_n^*(t) = \min_A f_{n,t}(A)$. Vertex switching and global coefficient negation preserve $f_{n,t}$. For $n \geq 3$ these transformations form a free group of size 2^n : global negation cannot be a switching, because switching multipliers have product $+1$ around every triangle. Therefore there are 2^{m-n} equal-weight orbit values, and

$$-\log f_n^*(t) - \frac{(m-n) \log 2}{q} \leq \frac{\mathcal{G}_n(q, t)}{q} \leq -\log f_n^*(t). \quad (15)$$

For an $F(n)$ -optimal signing, $Z_A(t) \leq 2^{n+1} e^{tF(n)}$, so

$$\log f_n^*(t) \leq tF(n) - m \log \cosh t.$$

Conversely, every signing has a maximizing projective pair $x, -x$. Choosing the favorable augmented sign for those two states gives $Z_A(t') \geq 2e^{t'F(n)}$, and hence

$$\log f_n^*(t') \geq t'F(n) - m \log \cosh t' - n \log 2.$$

Using these two estimates in (15) yields the exact finite bound

$$\Delta_n \geq q_{2n} \left[m(\log \cosh t_n - \log \cosh t_{2n}) - (t_n - t_{2n})F(n) - n \log 2 - \frac{(m-n) \log 2}{q_n} \right]. \quad (16)$$

Now $q_n/n \rightarrow \theta/\beta^2$, $q_{2n}/n \rightarrow 2\theta/\beta^2$, and

$$\frac{m}{n}(\log \cosh t_n - \log \cosh t_{2n}) \rightarrow \frac{\beta^2}{8}.$$

Taking the liminf proves (14). □

The right side is positive for sufficiently large θ whenever $\beta > 3.012373175204 \dots$; at $\beta = 4$ the threshold is $\theta > 7.690245425859 \dots$. The finite class-weighted values at $(\beta, \theta) = (4, 8)$ are

n	4	5	6	7	8	9
Δ_n/n^2	.57748	1.04741	1.17936	.99417	1.10086	1.11063.

They are regression evidence only; the theorem is analytic.

The gap also has an exact entropy-production mechanism.

Theorem 22 (Quantitative reverse-hypercontractive stability). *Let $m \geq 1$, let ν be uniform on $\{\pm 1\}^m$, and let $\theta > 0$. On any interval of nonnegative times for which $q_s = \theta e^{2s} - 1 > 0$, let*

$$L = \frac{1}{2} \sum_{i=1}^m (\tau_i - I), \quad f_s = T_{e^{-s}} f_0 > 0, \quad q_s + 1 = \theta e^{2s},$$

and put

$$\mu_s = \frac{f_s^{-q_s}}{\mathbb{E}_\nu f_s^{-q_s}} \nu, \quad D_s = D(\mu_s \| \nu), \quad H_s = \frac{1}{q_s} \log \mathbb{E}_\nu f_s^{-q_s}.$$

Then

$$\frac{dH_s}{ds} = -R_s, \quad R_s = \mathbb{E}_{\mu_s} \frac{L f_s}{f_s} - \frac{2(q_s + 1)}{q_s^2} D_s, \quad (17)$$

and

$$R_s \geq \frac{2D_s^2}{3q_s m}. \quad (18)$$

Equality holds only when f_s is constant.

Proof. Direct differentiation gives (17), using $q'_s = 2(q_s + 1)$ and

$$D_s = -q_s \mathbb{E}_{\mu_s} \log f_s - \log \mathbb{E}_\nu f_s^{-q_s}.$$

For one coordinate, condition on all other coordinates and call the two function values a, b . Put

$$z = \frac{q}{2} \left| \log \frac{b}{a} \right|, \quad d(z) = z \tanh z - \log \cosh z.$$

The conditional KL divergence is $d(z)$, while the conditional generator contribution is

$$e_q(z) = \frac{1}{2} \left(\frac{\cosh((1 + 2/q)z)}{\cosh z} - 1 \right).$$

If $r_q(z) = e_q(z) - 2(q + 1)d(z)/q^2$ and $c = 2/q$, then

$$q e_q(z) = \frac{\cosh(cz) - 1 + \tanh z \sinh(cz)}{c} \geq z \tanh z + c d(z),$$

where $d(z) \leq z^2/2$. Hence

$$q r_q(z) \geq 2 \log \cosh z - z \tanh z =: h(z).$$

Moreover $h(z) \geq 2d(z)^2/3$: the derivative of $h - 2d^2/3$ is nonnegative after using $d(z) \leq z^2/2$ and $\sinh z \cosh z \geq z + 2z^3/3$. Thus

$$r_q(z) \geq \frac{2d(z)^2}{3q}.$$

Average over the coordinate conditionals. If $\bar{d}_i = \mathbb{E} d_i$, entropy tensorization and Jensen–Cauchy give

$$D_s \leq \sum_i \bar{d}_i, \quad \sum_i \mathbb{E} d_i^2 \geq \sum_i \bar{d}_i^2 \geq \frac{(\sum_i \bar{d}_i)^2}{m} \geq \frac{D_s^2}{m}.$$

Moreover, exactly,

$$R_s = \sum_i \mathbb{E} r_q(z_i) + \frac{2(q + 1)}{q^2} \left(\sum_i \bar{d}_i - D_s \right).$$

The second term is the nonnegative dual total correlation. This proves (18). The scalar inequality is strict for $z > 0$, so equality forces every conditional z_i to vanish. Thus f_s is invariant under every coordinate flip and is constant. $\square$

For endpoints on the invariant curve,

$$\Delta_{n,N} = q_N \int_{s_n}^{s_N} R_s ds.$$

For proportional endpoints with fixed β, θ , one has $q_s \asymp n$, $m \asymp n^2$, and a fixed-length integration interval. Thus a power-saving transport defect would force $\int D_s^2 ds = O(n^{4-\delta})$. This quantifies scalar non-saturation, but does not recover the joint block geometry discarded by scalar transport.

12 Conditional alignment and the surviving block state

For a split $N = n + k$, let $R_{n,k}$ be the rectangular row-and-column switching code and put

$$C'_{n,k} = D_n \oplus D_k \oplus R_{n,k}.$$

For an $n \times k$ cross signing define

$$Z_C^\square(t) = \sum_{x \in \{-1,1\}^n, y \in \{-1,1\}^k} e^{tx^\top Cy}, \quad f_{n,k,t}^\square(C) = \frac{Z_C^\square(t)}{2^N (\cosh t)^{nk}},$$

and

$$\mathcal{G}_{n,k}^\square(q, t) = \log \mathbb{E}_C f_{n,k,t}^\square(C)^{-q}.$$

Theorem 23 (Exact conditional-alignment chain). *Suppose $n, k \geq 3$. Then $D_N \subset C'_{n,k}$,*

$$|C'_{n,k}/D_N| = 2^{N-1},$$

and the negative moment factors exactly as

$$\mathcal{G}_N(q, t) = \mathcal{G}_n(q, t) + \mathcal{G}_k(q, t) + \mathcal{G}_{n,k}^\square(q, t) + \mathcal{A}_{n,k}(q, t), \quad (19)$$

where

$$\mathcal{A}_{n,k}(q, t) = \log \mathbb{E}_{\hat{U}_q} K_q(h) \geq 0.$$

Here h is a coarse coset, r_h is its conditional fine density relative to uniform measure on the fiber, $K_q(h) = \mathbb{E}_{\text{fiber}} r_h^{-q}$, and $\hat{U}_q$ is the factorized coarse law tilted by its negative moment.

Proof. The dimensions are

$$\dim D_n = n, \quad \dim D_k = k, \quad \dim R_{n,k} = N - 1.$$

The three codes occupy disjoint edge-coordinate sets, so $\dim C'_{n,k} = 2N - 1$. Restricting every global cut and the global all-one word to the two internal blocks and the cross block proves $D_N \subset C'_{n,k}$. Since $\dim D_N = N$, the fiber size is 2^{N-1} . The coarse quotient factors into the two graph quotients and the rectangular quotient because the raw edge-noise law is a product.

For a coarse representative (A, B, C) , parameterize the fiber by

$$Y_{\alpha,\beta,\tau} = \begin{pmatrix} D_\alpha A D_\alpha & C \\ C^\top & \tau D_\beta B D_\beta \end{pmatrix},$$

where α, β are projective sign vectors and $\tau \in \{-1, 1\}$. These are the 2^{N-1} fine states. Direct changes of spin variables give

$$\mathbb{E}_{\alpha, \beta, \tau} Z_Y(t) = \frac{Z_A(t) Z_B(t) Z_C^\square(t)}{2^{N+1}}.$$

Thus

$$r_{A,B,C}(\alpha, \beta, \tau) = \frac{2^{N+1} Z_Y(t)}{Z_A(t) Z_B(t) Z_C^\square(t)}, \quad \mathbb{E}_{\text{fiber}} r = 1. \quad (20)$$

Grouping the uniform fine negative moment first by coarse fibers gives (19). Convexity of $z \mapsto z^{-q}$ gives $K_q(h) \geq 1$. $\square$

The powers of two in (20) use augmented graph state sums and the ordinary rectangular state sum. Codeword-sum conventions change the displayed constants.

Define

$$\mathcal{A}_{n,k}^\infty(t) = \lim_{q \rightarrow \infty} \frac{\mathcal{A}_{n,k}(q, t)}{q}.$$

If

$$h_n(t) = \min_A \log Z_A(t), \quad h_{n,k}^\square(t) = \min_C \log Z_C^\square(t),$$

then finite-space Laplace limits in (19) give exactly

$$\mathcal{A}_{n,k}^\infty(t) = h_n(t) + h_k(t) + h_{n,k}^\square(t) - (N+1) \log 2 - h_N(t).$$

Taking $t \rightarrow \infty$ sequentially therefore gives

$$\lim_{t \rightarrow \infty} \frac{\mathcal{A}_{n,k}^\infty(t)}{t} = F(n) + F(k) + B_\square(n, k) - F(N), \quad (21)$$

where

$$B_\square(n, k) = \min_C \max_{x,y} |x^\top C y|.$$

No exchange with $n \rightarrow \infty$ is asserted. Since $B_\square(n, n) \geq n \mathbb{E} |\varepsilon_1 + \dots + \varepsilon_n|$,

$$\liminf_n \frac{2F(n) + B_\square(n, n) - F(2n)}{n^{3/2}} \geq \frac{2}{\pi} + \sqrt{\frac{2}{\pi}} - \sqrt{2} = 0.0202907708\dots > 0. \quad (22)$$

Thus alignment is a leading ground-state quantity, not an auxiliary entropy correction.

The rectangular marginal cannot by itself repair the scalar transport gap. To state this precisely, put $\theta = \lambda \beta^2$ and

$$T_n(\beta, \lambda) = \frac{\mathcal{G}_n(q_n, t_n)}{q_n} - \frac{\mathcal{G}_n(q_{2n}, t_{2n})}{q_{2n}}.$$

Proposition 24 (Rectangular compensation is insufficient). *For fixed $\beta, \lambda > 0$,*

$$\begin{aligned} \limsup_{n \rightarrow \infty} \frac{1}{n} \left[\frac{\mathcal{G}_{n,n}^\square(q_{2n}, t_{2n})}{q_{2n}} - 2T_n(\beta, \lambda) \right] &\leq 4 \log 2 + \frac{\log 2}{\lambda} \\ &\quad - \beta \left[\frac{1}{\sqrt{\pi}} - \left(1 - \frac{1}{\sqrt{2}} \right) \right]. \end{aligned} \quad (23)$$

The coefficient of β is $0.2712963647\dots > 0$. First take $n \rightarrow \infty$ for each fixed (β, λ) as in the displayed limsup. Along a subsequent zero-temperature sequence with $\beta \rightarrow \infty$ and $\lambda \beta \rightarrow \infty$, the right side tends to $-\infty$.

Proof. For every $n \times n$ signing C , choose y uniformly and then set each x_i to the sign of $(Cy)_i$. This gives

$$\max_{x,y} |x^\top Cy| \geq \mathbb{E}_y \|Cy\|_1 = n\mu_n, \quad \mu_n = \mathbb{E}|\varepsilon_1 + \dots + \varepsilon_n|.$$

At $t_{2n} = \beta/\sqrt{2n}$, a maximizing projective pair contributes at least $2e^{t_{2n}n\mu_n}$ to every rectangular state sum. Therefore

$$\limsup_n \frac{1}{n} \frac{\mathcal{G}_{n,n}^\square(q_{2n}, t_{2n})}{q_{2n}} \leq \frac{\beta^2}{4} - \frac{\beta}{\sqrt{\pi}} + 2\log 2.$$

On the other hand, Theorem 21, with the audited $U \leq 1/2$ inserted and then divided by $q_{2n} \sim 2\lambda n$, gives

$$\liminf_n \frac{T_n}{n} \geq \frac{\beta^2}{8} - \frac{\beta}{2} \left(1 - \frac{1}{\sqrt{2}}\right) - \log 2 - \frac{\log 2}{2\lambda}.$$

Subtracting twice the second inequality from the first proves (23). $\square$

The first geometric component of this state is explicit.

Proposition 25 (Mixed four-cycle alignment Hamiltonian). *Put $P = D_\alpha A D_\alpha$ and $Q = D_\beta B D_\beta$. The coefficient of u^4 in $\log r_{A,B,C}(\alpha, \beta, \tau)$ is*

$$\mathcal{H}_4 = \frac{1}{2} \left[\text{tr}(P^2 C C^\top) + \text{tr}(Q^2 C^\top C) + \tau \text{tr}(P C Q C^\top) - nk(N-2) \right]. \quad (24)$$

It has the Walsh expansion

$$\begin{aligned} \mathcal{H}_4 = & \sum_{i < j} (A^2)_{ij} (C C^\top)_{ij} \alpha_i \alpha_j + \sum_{a < b} (B^2)_{ab} (C^\top C)_{ab} \beta_a \beta_b \\ & + \tau \sum_{i < j, a < b} a_{ij} b_{ab} (C_{ia} C_{jb} + C_{ib} C_{ja}) \alpha_i \alpha_j \beta_a \beta_b. \end{aligned} \quad (25)$$

Consequently

$$\begin{aligned} \mathbb{E} \mathcal{H}_4^2 = & \sum_{i < j} [(A^2)_{ij} (C C^\top)_{ij}]^2 + \sum_{a < b} [(B^2)_{ab} (C^\top C)_{ab}]^2 \\ & + \sum_{i < j, a < b} (C_{ia} C_{jb} + C_{ib} C_{ja})^2, \end{aligned} \quad (26)$$

and the last sum is at least

$$4 \binom{n}{2} \left\lfloor \frac{(k-1)^2}{4} \right\rfloor.$$

Proof. The graph and rectangular quotient duals have minimum weight four. Their signed weight-four sums are

$$c_4(S) = \frac{1}{8} [\text{tr} S^4 - v(v-1)(2v-3)]$$

and

$$r_4(C) = \frac{1}{4} [\text{tr}((C C^\top)^2) - nk(N-1)].$$

Direct block multiplication gives

$$\begin{aligned} \text{tr} Y^4 = & \text{tr} P^4 + \text{tr} Q^4 + 2 \text{tr}((C C^\top)^2) \\ & + 4 \text{tr}(P^2 C C^\top) + 4 \text{tr}(Q^2 C^\top C) + 4\tau \text{tr}(P C Q C^\top). \end{aligned}$$

Subtracting the three marginal coefficients proves (24). Expanding the switched entries and cancelling the diagonal baseline proves (25). Its characters are orthogonal, proving (26). For a fixed row pair, the final sum is four times the number of equal pairs among k signs, whose minimum is $\lfloor (k-1)^2/4 \rfloor$. $\square$

Thus $\mathcal{H}_4$ is nonconstant for $n, k \geq 3$. This does not justify truncating the Eulerian expansion on the mean-field diagonal: higher terms are not uniformly negligible when $u \asymp N^{-1/2}$ and $q \asymp N$.

Even complete local scalar free-energy curves do not determine alignment. At split $2+4$, in right-edge order $(01, 02, 03, 12, 13, 23)$, take

$$A = (1), \quad B = (-1, -1, -1, -1, -1, 1),$$

and

$$C_1 = \begin{pmatrix} -1 & -1 & -1 & -1 \\ -1 & -1 & 1 & 1 \end{pmatrix}, \quad C_2 = \begin{pmatrix} -1 & -1 & -1 & -1 \\ -1 & 1 & -1 & 1 \end{pmatrix}.$$

The two triples have identical complete local absolute-energy histograms. At $u = 3/5$, equivalently $t = \log 2$, exact rational evaluation gives

$$K_2(C_1) = \frac{196585091273040100817}{133610891512185651200}, \quad K_2(C_2) = \frac{6723290161}{5922841600}.$$

Their $\mathcal{H}_4$ laws are respectively

$$\{-12 : 4, -4 : 4, 0 : 16, 4 : 4, 12 : 4\}, \quad \{-4 : 8, 0 : 16, 4 : 8\}.$$

This small split is outside the $n, k \geq 3$ dimension statement in Theorem 23: $\dim D_2 = 1$. The displayed relative gauges cover the true 16-point fiber twice. Every density and $\mathcal{H}_4$ value is therefore repeated twice, leaving the normalized fiber moments, including the two values of K_2 , unchanged.

For clarity, fix $\beta, \theta > 0$, put $t_j = \beta/\sqrt{j}$, $q_j = \theta/\tanh^2(t_j) - 1$, and define

$$a_j = \frac{\mathcal{G}_j(q_j, t_j)}{q_j}, \quad T_{j \leftarrow N} = a_j - \frac{\mathcal{G}_j(q_N, t_N)}{q_N}.$$

The formal bound

$$\frac{\mathcal{G}_{n,k}^\square + \mathcal{A}_{n,k}}{q_N} \geq T_{n \leftarrow N} + T_{k \leftarrow N} - O(N^{1-\delta})$$

is not yet a reduction. The exact alignment chain gives

$$\frac{\mathcal{G}_{n,k}^\square + \mathcal{A}_{n,k}}{q_N} - T_{n \leftarrow N} - T_{k \leftarrow N} = a_N - a_n - a_k.$$

Thus the formal bound is precisely the desired almost-superadditivity in new notation. A useful next theorem must derive a summable bound from a smaller controlled state, such as a variational limit for the relative-switching Hamiltonian hierarchy.

There is, however, a non-tautological scalar relaxation of this hierarchy. It retains the complete near-maximal energy profiles of the three local blocks and discards only their additive labels.

13 A microcanonical relative-switching composition theorem

Write

$$\mathbb{P}_r = \{-1, 1\}^r / \{\pm \mathbf{1}\}, \quad G_{n,k} = \mathbb{P}_n \times \mathbb{P}_k \times \{-1, 1\}, \quad |G_{n,k}| = 2^{N-1}, \quad N = n + k.$$

For fixed signings A, B and a rectangular signing C , define

$$\Omega_A = \{-1, 1\} \times \mathbb{P}_n, \quad e_A(\sigma, [x]) = \sigma Q_A(x),$$

and define Ω_B, e_B analogously. Let

$$\overline{\mathcal{R}}_{n,k} = \{[xy^\top] : x \in \{-1, 1\}^n, y \in \{-1, 1\}^k\}, \quad e_C([R]) = |\langle C, R \rangle|,$$

where R and $-R$ are identified. Thus $|\Omega_A| = 2^n$, $|\Omega_B| = 2^k$, and $|\overline{\mathcal{R}}_{n,k}| = 2^{N-2}$.

There is a balanced homomorphism

$$\pi((\sigma, [z]), (\eta, [w]), [xy^\top]) = ([zx], [wy], \sigma\eta) \quad (27)$$

onto $G_{n,k}$. Its domain has 2^{2N-2} elements, so every fiber has 2^{N-1} elements.

Theorem 26 (Microcanonical relative-switching composition). *Put*

$$M_A = M(A), \quad M_B = M(B), \quad B_C = \max_{x,y} |x^\top C y|, \quad L = M_A + M_B + B_C,$$

and define local deficits

$$\begin{aligned} d_A(\sigma, [x]) &= M_A - \sigma Q_A(x), \\ d_B(\eta, [y]) &= M_B - \eta Q_B(y), \\ d_C([R]) &= B_C - |\langle C, R \rangle|. \end{aligned}$$

For $g = (\alpha, \beta, \tau) \in G_{n,k}$, let

$$Y_g = \begin{pmatrix} D_\alpha A D_\alpha & C \\ C^\top & \tau D_\beta B D_\beta \end{pmatrix}.$$

Then the exact max-plus identity

$$L - M(Y_g) = \min_{\pi(a,b,r)=g} [d_A(a) + d_B(b) + d_C(r)] \quad (28)$$

holds.

List all 2^{2N-2} product-triple deficit sums, with multiplicity, as

$$\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_{2^{2N-2}}, \quad \Lambda(A, B, C) = \lambda_{2^{N-1}}.$$

Then

$$F(N) \leq \min_g M(Y_g) \leq M(A) + M(B) + B_C - \Lambda(A, B, C). \quad (29)$$

Equivalently, if fewer than 2^{N-1} local triples have total deficit strictly below s , then some relative gauge satisfies $M(Y_g) \leq L - s$.

Proof. Fix a local triple. Choose a representative $R = xy^\top$ whose cross energy has the sign σ , and set $\alpha = [zx]$, $\beta = [wy]$, $\tau = \sigma\eta$. Evaluating Y_g at the full spin (x, y) and multiplying by σ gives exactly

$$e_A(\sigma, [z]) + e_B(\eta, [w]) + e_C([R]).$$

Thus the maximum local energy in a fiber is at most $M(Y_g)$. Conversely, every augmented full spin determines a triple in the corresponding fiber, and replacing its signed cross contribution by its absolute value can only increase the local sum. Hence

$$M(Y_g) = \max_{\pi(a,b,r)=g} [e_A(a) + e_B(b) + e_C(r)],$$

which is (28) after subtracting from L .

Choose one deficit-minimizing triple from each of the 2^{N-1} fibers. These triples are distinct, so the largest of their deficits is at least the 2^{N-1} -st global order statistic Λ . Equations (28) and (29) follow. $\square$

The theorem has a useful canonical scalar relaxation. For $t > 0$, put

$$D_A(t) = \sum_{a \in \Omega_A} e^{-td_A(a)}, \quad D_B(t) = \sum_{b \in \Omega_B} e^{-td_B(b)}, \quad D_C(t) = \sum_{r \in \mathcal{R}_{n,k}} e^{-td_C(r)}.$$

Exponential Markov counting gives

$$\Lambda(A, B, C) \geq \sup_{t>0} \left[\frac{(N-1) \log 2 - \log D_A(t) - \log D_B(t) - \log D_C(t)}{t} \right]_+. \quad (30)$$

This is not the failed weighted Hamming-ball union bound. It does not randomize C or sum one-point Hamming bad events. It counts the actual simultaneously near-maximal local states and uses their distribution among the relative gauges. There is no known analogue here of the universal $2\sqrt{\log 2}$ leading floor.

The order-statistic guarantee is the strongest possible statement that uses only the unlabeled multiset of product-triple deficits and the equal fiber sizes. Indeed, place the 2^{N-1} smallest deficits one per fiber and then fill each fiber with no-smaller remaining values. The largest fiber minimum is exactly Λ . Additive labels can make the true gain strictly larger. For the two $2+4$ scalar-collision examples above, both unlabeled profiles have $\Lambda = 0$, while their exact gains $L - \min_g M(Y_g)$ are respectively 4 and 2.

At finite temperature, the same map identifies the conditional alignment law as the pushforward convolution of the two augmented graph Gibbs laws and the rectangular law with weight $2 \cosh(t\langle C, R \rangle)$. Its Fourier coefficients factor into three local correlation tensors. For even $I \subseteq [n]$, even $J \subseteq [k]$, and $\epsilon \in \{0, 1\}$,

$$\hat{r}(I, J, \epsilon) = \mathbb{E}_{\mu_A}[\sigma^\epsilon z_I] \mathbb{E}_{\mu_B}[\eta^\epsilon w_J] \mathbb{E}_{\mu_C}[x_I y_J]. \quad (31)$$

Indeed, the character indexed by (I, J, ϵ) pulls back under π to $\sigma^\epsilon z_I \eta^\epsilon w_J x_I y_J$ on the three product factors, so independence gives the displayed product. Conditional expectation onto a character subspace preserves normalization, and Jensen's inequality for $z \mapsto z^{-q}$ shows that projection can only decrease $\mathbb{E}r^{-q}$. Equation (31) therefore supplies an auditable projected hierarchy between the scalar order statistic and the full alignment state; the mixed terms in Proposition 25 are its first nonzero layer.

The exact scalar settling target is now explicit. A composable near-optimal choice of A, B, C would suffice if, for some $\delta > 0$, the number of local triples below the deficit threshold

$$M(A) + M(B) + B_C - [F(n)^{2/3} + F(k)^{2/3}]^{3/2} - O(N^{3/2-\delta})$$

were less than 2^{N-1} . Theorem 26 would then give power-saving near-subadditivity of $F^{2/3}$. Constantly many exact maximizers do not by themselves prove this: control of the entire leading-scale near-maximal profile is required. If this scalar quantile is too small, the strict collision gains show exactly where the additive Fourier geometry can still recover information.

There is an equivalent cumulant formulation that cleanly separates the live large-deviation problem from two insufficient shortcuts. Let (a, b, c) be uniform on the three local product spaces and put

$$X = e_A(a) + e_B(b) + e_C(c).$$

Proposition 27 (Exact scalar cumulant criterion). *For every $t > 0$,*

$$\min_g M(Y_g) \leq \frac{(N-1) \log 2 + \log \mathbb{E} e^{tX}}{t}. \quad (32)$$

Equivalently, if

$$\kappa_N(\beta) = \frac{1}{N} \log \mathbb{E} \exp \left(\frac{\beta X}{\sqrt{N}} \right),$$

then a threshold $\theta_N N^{3/2}$ is certified whenever

$$\kappa_N(\beta) - \beta \theta_N \leq - \left(1 - \frac{1}{N} \right) \log 2. \quad (33)$$

This criterion is exactly the exponential relaxation (30), not a new loss.

Proof. The product space has 2^{2N-2} points. Strict exponential Markov counting gives

$$\#\{X > T\} < 2^{2N-2} e^{-tT} \mathbb{E} e^{tX}.$$

If $tT \geq (N-1) \log 2 + \log \mathbb{E} e^{tX}$, fewer than 2^{N-1} triples exceed T . Since there are 2^{N-1} equal fibers, one fiber has no such triple, and its relative gauge has maximum at most T . Finally,

$$\mathbb{E} e^{tX} = 2^{-(2N-2)} e^{tL} D_A(t) D_B(t) D_C(t),$$

which proves the last assertion and the mean-field rescaling. $\square$

The full cumulant function may still contain the needed tail information. A quadratic proxy provably does not.

Proposition 28 (Quadratic-cumulant certificate barrier). *Take balanced blocks $n = k = r$. Suppose one bounds the centered local sum by*

$$\log \mathbb{E} e^{t(X - \mathbb{E}X)} \leq \frac{v_r t^2}{2} \quad (t > 0).$$

Every such bound has $v_r \geq r(r-1)$. Consequently, optimizing (32) with this quadratic majorant gives a certified threshold whose normalized asymptotic constant is at least

$$\sqrt{\frac{\log 2}{2}} = 0.588705011 \dots > \frac{1}{2}. \quad (34)$$

This is a no-go theorem only for Gaussian or quadratic-cumulant certificates; it is not a lower bound on the actual order statistic Λ .

Proof. The augmented graph energies have mean zero, and Walsh orthogonality gives

$$\text{Var}(e_A) = \text{Var}(e_B) = \binom{r}{2}.$$

The three local states are independent, so $\text{Var}(X) \geq r(r-1)$. Comparison at the second derivative at zero forces $v_r \geq \text{Var}(X)$. Also $0 \leq \mathbb{E}X = \mathbb{E}|x^\top Cy| \leq r$. The optimized quadratic certificate is

$$\mathbb{E}X + \sqrt{2v_r(2r-1)\log 2}.$$

After division by $(2r)^{3/2}$, its second term has lower limit (34), while the mean is negligible. $\square$

Ordinary degree-two hypercontractivity is weaker at the required tail scale. Indeed,

$$\|Q_A\|_p \leq (p-1)\|Q_A\|_2 = (p-1)\sqrt{\binom{n}{2}}, \quad p \geq 2.$$

Optimized at a threshold $cn^{3/2}$, this yields only $\exp[-\Theta_c(\sqrt{n})]$, whereas Theorem 26 needs control at probability $2^{-\Theta(n)}$. Thus this standard hypercontractive moment certificate does not resolve the microcanonical threshold.

For completeness, there is also a structural operator-norm estimate. It is rigorous but quantitatively too weak to close the profile criterion.

Proposition 29 (PSD-shift moment bound). *Let A be a symmetric zero-diagonal signing of order $n \geq 2$ and $\rho = \|A\|_{\text{op}}$. For $0 < t < (2\rho)^{-1}$,*

$$\log \mathbb{E}_x e^{tQ_A(x)} \leq \frac{t^2\{n(n-1) + n\rho^2\}}{4(1-2t\rho)}. \quad (35)$$

The same estimate holds with $Q_A(x)$ replaced by the augmented variable $\sigma Q_A(x)$. If C is an $n \times k$ sign matrix, $N = n + k$, and $\rho_C = \|C\|_{\text{op}}$, then, for $0 < t < (2\rho_C)^{-1}$,

$$\log \mathbb{E}_{x,y} e^{t|x^\top Cy|} \leq \log 2 + \frac{t^2\{2nk + N\rho_C^2\}}{4(1-2t\rho_C)}. \quad (36)$$

Proof. Put $B = A + \rho I \succeq 0$. Gaussian linearization followed by $\cosh u \leq e^{u^2/2}$ gives

$$\mathbb{E}_x e^{tQ_A(x)} \leq e^{-t\rho n/2} \det(I - tB)^{-1/2}.$$

If $b_i \in [0, 2\rho]$ are the eigenvalues of B , the logarithm of the right side is

$$\frac{1}{2} \sum_i [-\log(1 - tb_i) - tb_i] \leq \frac{t^2 \sum_i b_i^2}{4(1-2t\rho)}.$$

Now $\sum_i b_i^2 = n(n-1) + n\rho^2$, proving (35). Applying it also to $-A$ and averaging proves the augmented assertion. Apply the same argument to $\begin{pmatrix} 0 & C \\ C^\top & 0 \end{pmatrix}$ and use $e^{t|z|} \leq e^{tz} + e^{-tz}$ to obtain (36). $\square$

Exact finite calibration shows both what the scalar theorem captures and what it currently misses. The verifier exhausts every balanced split of the two stored rooted order-12 survivor encodings, one stored order-13 Paley principal witness, and the order-14 Paley witness. In the table, $G = L - F(N)$ is the exact total gain, E_* is the smallest energy of the correct parity strictly above

$[F(n)^{2/3} + F(k)^{2/3}]^{3/2}$, and the last column counts product triples with energy at least E_* against the fiber count 2^{N-1} .

N	split	# splits	Λ	$L - \Lambda$	$G - \Lambda$	$E_* : \# / 2^{N-1}$
12	6 + 6	462 each	0 ... 10	24 ... 26	6 ... 8	16 : 213312 ... 269122 / 2048
13	6 + 7	1716	2 ... 10	28 ... 30	8 ... 10	20 : 436864 ... 495632 / 4096
14	7 + 7	1716	6 ... 10	31 ... 33	10 ... 12	27 : 231581 ... 305465 / 8192

Thus these particular scalar profiles do not certify the exact near-subadditive target: even the smallest target count exceeds one fiber by a factor greater than 28. The additive labels recover a further 6–12 energy units. This is finite evidence, not an asymptotic no-go: the best scalar excesses above the target are only 9.858..., 8.476..., and 5.544..., respectively, and an $O(N)$ composition error would be a valid power saving at the $N^{3/2}$ scale.

The state-space convention in Theorem 26 is essential for its exact max-plus identity. There is, however, a second valid one-sided profile theorem. Put

$$\tilde{\Omega} = \mathbb{P}_n \times \mathbb{P}_k \times \{-1, 1\}^n \times \{-1, 1\}^k, \quad \tilde{e}([z], [w], x, y) = |Q_A(z)| + |Q_B(w)| + x^\top C y,$$

and define

$$\tilde{\pi}([z], [w], x, y) = ([zx], [wy], x_1).$$

This map is onto $G_{n,k}$ and has 2^{N-1} states in each fiber.

Proposition 30 (One-sided swapped-profile domination). *For every $g \in G_{n,k}$,*

$$\max_{\tilde{\pi}(\omega)=g} \tilde{e}(\omega) \geq M(Y_g).$$

If $\tilde{\Lambda}$ is the 2^{N-1} -st smallest deficit $L - \tilde{e}(\omega)$, with multiplicity, then

$$F(N) \leq \min_g M(Y_g) \leq L - \tilde{\Lambda}.$$

Proof. Write $g = (\alpha, \beta, \tau)$ and choose u, v and $s \in \{-1, 1\}$ such that $sQ_{Y_g}(u, v) = M(Y_g)$. Set $z = [\alpha u]$, $w = [\beta v]$, $r = \tau / (su_1)$, $x = rsu$, and $y = rv$. Then $\tilde{\pi}([z], [w], x, y) = g$, while

$$\tilde{e}([z], [w], x, y) = |Q_A(\alpha u)| + |Q_B(\beta v)| + su^\top C v \geq sQ_{Y_g}(u, v).$$

Choosing one such state in each fiber gives 2^{N-1} distinct deficits at most $L - \min_h M(Y_h)$, proving the order-statistic claim. $\square$

This domination need not be an equality, so it is not a verification of Theorem 26. On the standard balanced C_{14} split it gives $\tilde{\Lambda} = 8$ and 596440 target states. The exact theorem gives $\Lambda = 10$ and 304908 target states; the true gain is 22. Both scalar bounds are valid, and neither finite target count rules out an asymptotically acceptable $O(N)$ defect. The finite verifiers distinguish the two maps.

The alternate raw statistic has a universal floor. In a balanced $r + r$ split, put $\mu_r = \mathbb{E}|\varepsilon_1 + \dots + \varepsilon_r|$ and let $\tilde{U} = L - \tilde{\Lambda}$ be its order-statistic energy bound. Then

$$\tilde{U} \geq \max\{M(A), M(B)\} + r\mu_r - 3r. \quad (37)$$

Indeed, $H(y) = \|Cy\|_1$ has mean $r\mu_r$ and is at most r^2 , so at least $2^r / (r + 1)$ full spins satisfy $H(y) \geq r\mu_r - r$. For each such y , its maximizing row spin and the r one-coordinate flips give $r + 1$

distinct signed pairs with energy at least $H(y) - 2r$. There are therefore at least 2^r cross states above $r\mu_r - 3r$. Combine them with one maximizing projective state of A and every projective state of B ; the latter absolute energies are nonnegative. This reaches the rank 2^{2r-1} . Interchanging the blocks proves (37).

Consequently, if this raw profile alone gave $\tilde{U} \leq 2^{3/2}F(r) + o(r^{3/2})$, then

$$\liminf_{r \rightarrow \infty} \frac{F(r)}{r^{3/2}} \geq \frac{\sqrt{2/\pi}}{2^{3/2} - 1} = 0.4363775564 \dots \quad (38)$$

This is compatible with a limit of $1/2$ and does not obstruct the distinct exact order statistic in Theorem 26.

The canonical exponential relaxations of the two profiles are much closer. Let $P_{\text{aug}}(t)$ and $P_{\text{sw}}(t)$ be the uniform product moment generating functions of their local energy sums, put $H = (N-1)\log 2$, and define

$$U_* = \inf_{t>0} \frac{H + \log P_*(t)}{t}.$$

Pointwise, the two graph-factor comparisons $e^{t|q|}/2 \leq \cosh(tq) \leq e^{t|q|}$ and the reverse rectangular comparison give

$$\frac{1}{2} \leq \frac{P_{\text{sw}}(t)}{P_{\text{aug}}(t)} \leq 4.$$

Since both moment generating functions are at least one and both optimized bounds are at most L , taking approximate minimizers gives exactly

$$U_{\text{aug}} - \frac{L}{N-1} \leq U_{\text{sw}} \leq U_{\text{aug}} + \frac{2L}{N-1}. \quad (39)$$

For example, at an approximate augmented minimizer t , $H/t \leq L + o(1)$ and the pointwise upper loss is $(\log 4)/t$; the other direction is identical. Thus the canonical bounds differ by only $O(\sqrt{N})$ when $L = O(N^{3/2})$. Combining this with (37) transfers the conditional constant barrier (38) to the full canonical exponential relaxation of either convention. It does not transfer to the raw exact augmented order statistic.

The additive labels admit a strict deterministic refinement of the scalar order statistic. For a threshold s , define

$$b_s(g) = \#\{(a, b, r) : \pi(a, b, r) = g, d_A(a) + d_B(b) + d_C(r) < s\}, \quad K = |G_{n,k}| = 2^{N-1}.$$

Proposition 31 (Labeled-shell Parseval bonus). *Put*

$$\mu_s = K^{-1} \sum_g b_s(g), \quad V_s = \sum_{\chi \neq 1} |\hat{b}_s(\chi)|^2 = K^{-1} \sum_g (b_s(g) - \mu_s)^2,$$

where $\hat{b}_s(\chi) = K^{-1} \sum_g b_s(g) \chi(g)$. Then

$$\min_g b_s(g) \leq \mu_s - \sqrt{\frac{V_s}{K-1}}. \quad (40)$$

If the right side is less than one, some fiber is empty and $\min_g M(Y_g) \leq L - s$.

Proof. Let $h = b_s - \mu_s$ and $m = -\min h$. Since $h \geq -m$ and $\sum_g h(g) = 0$, one also has $h \leq (K-1)m$. Therefore $(h+m)(h-(K-1)m) \leq 0$. Averaging and using $\mathbb{E}h = 0$ gives $V_s \leq (K-1)m^2$, proving (40). If its right side is below one, the nonnegative integer $\min b_s$ is zero. The max-plus identity (28) then gives the asserted energy bound. $\square$

The coefficient computation retains only local labeled shells. For even $I \subseteq [n]$, even $J \subseteq [k]$, and $\epsilon \in \{0, 1\}$, set

$$\begin{aligned} A_d(I, \epsilon) &= \sum_{d_A(\sigma, [z])=d} \sigma^\epsilon z_I, & B_d(J, \epsilon) &= \sum_{d_B(\eta, [w])=d} \eta^\epsilon w_J, \\ C_d(I, J) &= \sum_{d_C([xy^\top])=d} x_I y_J. \end{aligned}$$

Pullback of the character (I, J, ϵ) under π gives

$$\widehat{b}_s(I, J, \epsilon) = \frac{1}{K} \sum_{d_A+d_B+d_C=s} A_{d_A}(I, \epsilon) B_{d_B}(J, \epsilon) C_{d_C}(I, J). \quad (41)$$

Thus Proposition 31 is a computable labeled improvement, not a restatement of the unknown fiber minimum.

At threshold $s = 2$ in the exact split-2 + 4 collision, precisely $K = 32$ product triples have deficit zero. The scalar criterion gives no gain, but the fiber occupancy law is $0^8, 1^{16}, 2^8$. Hence $\mu_s = 1$, $V_s = 1/2$, and $\min b_s \leq 1 - 1/\sqrt{62} < 1$, certifying gain two. The two actual gains are four and two. For the standard balanced C_{14} split at target energy 27 (equivalently $s = 18$), the corresponding values are

$$K = 8192, \quad \sum_g b_s(g) = 304908, \quad V_s = 151.4690835 \dots, \quad \min b_s = 0, \quad \max b_s = 87.$$

There are 8159 nonzero nontrivial coefficients and exactly one empty fiber among 8192, but (40) gives only $37.0842 \dots > 1$. The variance bound is therefore strict and useful at the scalar boundary, but a higher-moment or lower-tail theorem is needed in the dense target regime.

For integer occupancies there is a complete moment hierarchy. If $b : G \rightarrow \mathbb{Z}_{\geq 0}$ and $A \subset \mathbb{Z}_{\geq 1}$ is finite, put

$$P_A(x) = (1 - x) \prod_{a \in A} \frac{(x - a)(x - a - 1)}{a(a + 1)}.$$

Then $P_A(0) = 1$ and $P_A(j) \leq 0$ at every positive integer j .

Proposition 32 (Integer-moment vacancy certificate). *For every such A ,*

$$\frac{\#\{g : b(g) = 0\}}{|G|} \geq \mathbb{E}_g P_A(b(g)). \quad (42)$$

If $m_j = \mathbb{E} b^j$ and $H_r = (m_{i+j+1} - m_{i+j})_{0 \leq i, j \leq r}$, an everywhere-positive occupancy forces $H_r \succeq 0$. This hierarchy is complete: if $t_1, \dots, t_u$ are the distinct positive values of b , then $q(x) = \prod_i (x - t_i)$ gives $\mathbb{E}[(b - 1)q(b)^2] < 0$ whenever a zero exists. Moreover,

$$m_j = \sum_{\chi_1 \cdots \chi_j = 1} \widehat{b}(\chi_1) \cdots \widehat{b}(\chi_j). \quad (43)$$

Proof. For integer j , every factor $(j - a)(j - a - 1)$ is nonnegative. This proves the sign assertion for P_A and hence (42). If $b \geq 1$, then for every $q(x) = \sum_{i=0}^r c_i x^i$,

$$c^\top H_r c = \mathbb{E}[(b - 1)q(b)^2] \geq 0.$$

The displayed choice of q proves completeness. Finally, expand $b(g) = \sum_\chi \widehat{b}(\chi) \chi(g)$ and average; character orthogonality leaves exactly the products with $\chi_1 \cdots \chi_j = 1$. $\square$

If the values lie in $\{0\} \cup [m, R]$ with $0 < m < R$ and a zero exists, the hierarchy has a generic quantitative bound. Map $[m, R]$ to $[-1, 1]$ and normalize the Chebyshev polynomial T_r to equal one at zero. The resulting localizing form is negative whenever

$$T_r\left(\frac{R+m}{R-m}\right)^2 > |G|(R-1).$$

The zero contributes at most $-|G|^{-1}$, while the total positive contribution is bounded by $(R-1)$ divided by the left-side Chebyshev square. For the C_{14} values $|G| = 8192, m = 6, R = 87$, this generic test first succeeds at $r = 14$; the tailored witness below succeeds at $r = 9$.

For the balanced C_{14} shell, take

$$A = \{9, 17, 26, 36, 46, 56, 67, 76, 86\}.$$

The direct exact calculation is

$$\mathbb{E}P_A(b) = \frac{1707454816960049615}{99244391564512637853696} > 0. \quad (44)$$

Thus the first 19 moments certify an empty fiber. Equivalently, the degree-nine polynomial $q(x) = \prod_{a \in A} (2x - (2a + 1))$ satisfies

$$\sum_g (1 - b(g))q(b(g))^2 = 584163517696745929254421003286532 > 0.$$

The hierarchy does not give a bounded-order asymptotic theorem for free. For $D \geq 1$, form one multiset from each even j with multiplicity $\binom{D+1}{j}$ and another from the odd j with the same multiplicity. Both have size 2^D ; the even multiset has one zero and the odd multiset has none. Their moments through degree D agree by

$$\sum_j (-1)^j \binom{D+1}{j} j^r = 0 \quad (0 \leq r \leq D).$$

Thus generic moments through degree $N-1$ cannot decide vacancy on 2^{N-1} fibers. This abstract obstruction need not be realizable by signing shells, but it shows that the missing estimate must exploit the factorization (41) at degree growing with N . Indeed, the character matrix $\mathcal{H}_{\chi, \psi} = \hat{b}(\chi\psi)$ has eigenvalues $b(g)$ and is positive semidefinite. If there is exactly one zero and all positive values are at least m , Cauchy–Schwarz shows that a negative witness for $\mathcal{H} - I$ supported on q characters must satisfy $q > |G|(m-1)/m$: writing $p(g) = \sum_{\chi} c_{\chi} \chi(g)$, negativity gives $(m-1)|G|\|c\|_2^2 < m|p(g_0)|^2 \leq mq\|c\|_2^2$. For the calibrated C_{14} shell, $m = 6$, so this requires at least 6827 of 8192 character labels. Fixed-order mixed traces and sparse Fourier support are therefore insufficient generic substitutes.

The fixed-half-density reformulation also permits an exact audit of the rectangular block. Let $C \subseteq U \times V$, with $|U| = n$, $|V| = k$, be encoded by $S \in \{-1, 1\}^{n \times k}$ via $S_{ij} = 2\mathbf{1}_C(i, j) - 1$, and put $t = \sum_{ij} S_{ij}$ and $\|S\|_{\infty \rightarrow 1} = \max_{x \in \{-1, 1\}^n, y \in \{-1, 1\}^k} |x^T S y|$. Define

$$D_{\oplus}(C) = \max_{P \subseteq U, R \subseteq V} \left| e_C(P, V \setminus R) + e_C(U \setminus P, R) - \frac{|P|(k - |R|) + (n - |P|)|R|}{2} \right|.$$

Proposition 33 (Fixed-density rectangular cross floor). *One has*

$$D_{\oplus}(C) = \frac{|t| + \|S\|_{\infty \rightarrow 1}}{4}, \quad \|S\|_{\infty \rightarrow 1} \geq \max\{n\mu_k, k\mu_n\}, \quad (45)$$

where $\mu_j = \mathbb{E}|\varepsilon_1 + \dots + \varepsilon_j|$. If

$$R_{n,k}(t_0) = \min_{\sum S_{ij}=t_0} \|S\|_{\infty \rightarrow 1}, \quad R_{n,k} = \min_S \|S\|_{\infty \rightarrow 1},$$

for an admissible $t_0 \equiv nk \pmod{2}$, then

$$R_{n,k} \leq R_{n,k}(t_0) \leq R_{n,k} + \sqrt{nk} + |t_0|. \quad (46)$$

Proof. Membership signs x, y turn the quantity inside the absolute value into $(t - x^\top Sy)/4$. Replacing x by $-x$ proves the first identity. For uniform y , coordinatewise optimization gives $\max_x x^\top Sy = \|Sy\|_1$ and $\mathbb{E}_y \|Sy\|_1 = n\mu_k$; transpose for the other bound.

For (46), row and column switching preserves the norm and satisfies $\mathbb{E}_{x,y} (x^\top Sy)^2 = nk$. Some switch of an unrestricted optimizer therefore has total sum at most $\sqrt{nk}$ in absolute value. Flipping $|t - t_0|/2$ entries reaches total t_0 and changes the norm by at most $|t - t_0|$. $\square$

When two fixed-half internal blocks are joined inside the global fixed-half layer, $|t_0| \leq 2$. Thus imposing the density costs only $O(\sqrt{nk})$, but for $n = k = r$ the separate cross deviation still has the leading floor

$$D_\oplus(C) \geq \left(\frac{\sqrt{2/\pi}}{4} + o(1) \right) r^{3/2}.$$

This rules out only triangle-inequality composition that controls the cross block independently. Labeled cancellation with the internal blocks remains the live mechanism.

The fixed-half formulation also makes the failure of ordinary equal-cloud amplification exact. For a graph G , write

$$d(G) = \max_{U \subseteq V(G)} \left| e_G(U, U^c) - \frac{1}{2} |U| |U^c| \right|.$$

Proposition 34 (Fixed-half equal-cloud blow-up obstruction). *Let G be fixed-half on $n \geq 3$ vertices and replace each vertex by a cloud of size k . Make every pair of clouds complete or empty according to the corresponding edge of G , and choose the within-cloud edges so that the resulting graph L_0 is fixed-half. Such a choice always exists, and*

$$d(L_0) \geq k^2 d(G).$$

If L_1 is any Seidel switch of L_0 and a fixed-half graph L' is obtained from L_1 by toggling r edges, then, with $t_N = \binom{nk}{2} \pmod{2}$,

$$d(L') \geq k^2 d(G) - \frac{r + t_N}{2}. \quad (47)$$

For $n = 2$ and $k > 1$, the asserted fixed-half completion can fail.

Proof. For any equal-cloud graph put $\beta_{ij} = e(V_i, V_j) - k^2/2$. A cloud-union cut gives

$$d(L_0) \geq \max_{S \subseteq [n]} \left| \sum_{i \in S, j \notin S} \beta_{ij} \right|.$$

In the complete/empty construction the right side is $k^2 d(G)$.

For feasibility, encode edges by $+1$ and nonedges by -1 , put $m_j = \binom{j}{2}$ and $t_j = m_j \pmod{2}$. The off-cloud sign total is $-k^2 t_n$, while the target total is $-t_{nk}$. Thus the nm_k internal signs must have total $k^2 t_n - t_{nk}$. Its parity is correct by $m_{nk} = k^2 m_n + nm_k$, and its absolute value is at most

nm_k for $n \geq 3$; if $t_n = 0$ this is immediate. If $t_n = 1$, then $k = 1$ is trivial; for $n = 3$, the cases $k = 2, 3$ are tight and $k \geq 4$ follows from $k^2 \leq 3k(k-1)/2$; the next possible n is at least six, where $k^2 \leq nk(k-1)/2$ for $k \geq 2$. Hence the internal signs can be chosen.

For a fixed-half graph of order N , its signing maximum M satisfies $|M - 4d| \leq t_N$. Seidel switching preserves M , while one edge toggle changes it by at most two. Therefore

$$4d(L') \geq M(L') - t_N \geq M(L_0) - 2r - t_N \geq 4d(L_0) - 2r - 2t_N,$$

which proves (47). $\square$

Thus $O(nk)$ density repair cannot remove the $\sqrt{k}$ normalized loss of a complete/empty blow-up. Orthogonal blocks evade this particular cloud-union estimate because a Hadamard matrix can have $|\mathbf{1}^\top H \mathbf{1}| = O(k^{3/2})$, or even zero. Fine cuts still defeat the smallest common-block attempt. Let A be the order-five signing with negative cycle edges and positive chords. For every symmetric 4×4 Hadamard matrix H and every fixed-half order-four signing D , the uniform lift

$$S = A \otimes H + I_5 \otimes D$$

is fixed-half of order 20 and satisfies

$$M(S) \geq 44 > 4^{3/2}M(A) = 32.$$

Equality occurs. The exact verifier classifies all 64 symmetric Hadamard matrices. Every such H is $sEPH_i P^\top E$ for a diagonal sign matrix E and one of the two tested representatives. The signed permutation is absorbed by a common relabeling and switching inside each cloud; if $s = -1$, the base permutation $(0, 2, 4, 1, 3)$ sends A to $-A$. The verifier checks this anti-isomorphism, exhausts all transformed diagonal completions and all 2^{19} projective full spins, and exhibits both the value-44 signing and a fine-spin witness. This is a finite no-go for the uniform 4-fold family only, not for general orthogonal or cloud-dependent lifts.

14 Conference alignment as an ℓ_1 -eigenspace problem

For every symmetric conference matrix, Boolean spectral alignment has an exact geometric form that treats both signs of the quadratic form.

Theorem 35 (Conference eigenspace formula). *Let C be a symmetric conference matrix of order n , put $r = \sqrt{n-1}$, and let*

$$E_\pm = \ker(C \mp rI), \quad \alpha_\pm = \frac{1}{\sqrt{n}} \sup_{\substack{v \in E_\pm \\ \|v\|_2=1}} \|v\|_1.$$

Then

$$\frac{2M(C)}{n\sqrt{n-1}} = \max\{2\alpha_+^2 - 1, 2\alpha_-^2 - 1\}. \quad (48)$$

In particular, along a sequence, asymptotic spectral saturation is equivalent to at least one eigenspace containing unit vectors whose ℓ_1 norm is $(1 - o(1))\sqrt{n}$. If instead both $\alpha_\pm^2 \leq 1 - \eta$, then

$$M(C) \leq \left(\frac{1}{2} - \eta\right) n\sqrt{n-1}.$$

Proof. For an orthogonal projection P onto a subspace E ,

$$\max_{x \in \{-1,1\}^n} x^\top P x = \left(\sup_{\substack{v \in E \\ \|v\|_2=1}} \|v\|_1 \right)^2. \quad (49)$$

Indeed, $\|Px\|_2 = \sup_{v \in E, \|v\|_2=1} \langle v, x \rangle$, and maximizing over x changes the inner supremum into $\|v\|_1$. For $P_\pm = (I \pm C/r)/2$, trace zero gives $\text{tr } P_\pm = n/2$, so $\alpha_\pm^2 \geq 1/2$. Since $C = r(2P_+ - I)$ and $-C = r(2P_- - I)$, applying (49) to the positive and negative maxima proves (48). Cauchy gives $\alpha_\pm \leq 1$ and the remaining claims follow. Quantitatively, if $s_i = \text{sign}(v_i)$, then

$$\left\| v - \frac{s}{\sqrt{n}} \right\|_2^2 = 2 \left(1 - \frac{\|v\|_1}{\sqrt{n}} \right),$$

so $\alpha \rightarrow 1$ is precisely approach to a Boolean cube direction. $\square$

Formula (48) identifies a concrete possible obstruction to universal conference alignment: the two eigenspaces could be uniformly Euclidean in the sense that both ℓ_1 suprema stay below $(1-\eta)^{1/2}\sqrt{n}$. No theorem used here rules this out for arbitrary conference matrices. The Paley results below exhibit special eigenspaces that do approach cube directions.

15 A known square-order Paley ceiling theorem

The following classical regular-conference construction corrects an overstated external report. It is not a new result [11, 12].

Theorem 36 (Regular Paley Boolean eigenvector). *Let m be an odd prime power. A Paley conference matrix C of order $m^2 + 1$ has a Boolean eigenvector of eigenvalue m , and*

$$M(C) = \frac{m(m^2 + 1)}{2}.$$

Proof. Let $K = \mathbb{F}_{m^2}$, $H = \mathbb{F}_m$, and let χ be the quadratic character of K . Index C by $\{\infty\} \cup K$, with $C_{\infty,a} = 1$ and $C_{a,b} = \chi(a - b)$. Choose signs $\sigma : K/H \rightarrow \{-1, 1\}$ with $(m+1)/2$ positive cosets, so $\sum_c \sigma(c) = 1$, and put $x_\infty = 1$, $x_a = \sigma(a + H)$.

Every element of H^* is a square in K , so the character sum on the zero additive coset is $m - 1$. Multiplication by H^* is transitive on the nonzero cosets of the one-dimensional quotient K/H and preserves χ . All nonzero-coset sums are therefore equal; the total character sum on K is zero, so each equals -1 . For a in coset c ,

$$(Cx)_a = 1 + (m-1)\sigma(c) - \sum_{d \neq c} \sigma(d) = m\sigma(c), \quad (Cx)_\infty = m.$$

Thus $Cx = mx$. The conference identity $C^2 = m^2 I$ supplies the matching spectral upper bound. $\square$

These full Paley matrices have normalized Boolean maximum tending to $1/2$. This gives no lower bound on the minimax $F(m^2 + 1)$. Already $m = 3$ gives $M(C) = 15$ at order ten, while $F(10) = 13$. The theorem neither improves the known upper bound nor excludes better signings or principal submatrices.

16 Paley alignment as Fourier leakage

Beyond the square- q Paley subfamily, the Boolean-versus-spectral alignment question has an exact additive-Fourier form. The identity below is a reformulation for one explicit matrix family, not a bound on the minimax F .

Proposition 37 (Paley Fourier-leakage identity). *Let $q \equiv 1 \pmod{4}$ be an odd prime power, let C be the Paley conference matrix of order $q + 1$, and fix a nontrivial additive character ψ of $\mathbb{F}_q$. If χ is the quadratic character, write*

$$\tau = \sum_{a \in \mathbb{F}_q} \chi(a) \overline{\psi(a)} = \epsilon_q \sqrt{q}, \quad \epsilon_q \in \{-1, 1\},$$

and put

$$H_{\pm} = \{r \in \mathbb{F}_q^* : \epsilon_q \chi(r) = \pm 1\}.$$

For $f : \mathbb{F}_q \rightarrow \{-1, 1\}$, define the unitary transform

$$\widehat{f}(r) = q^{-1/2} \sum_{a \in \mathbb{F}_q} f(a) \overline{\psi(ra)}, \quad S(f) = \sum_a f(a), \quad E_{\pm}(f) = \sum_{r \in H_{\pm}} |\widehat{f}(r)|^2,$$

and $W(f) = \min(E_+(f), E_-(f))$. Then

$$M(C) = \frac{(q+1)\sqrt{q}}{2} - \sqrt{q} \min_{f: \mathbb{F}_q \rightarrow \{-1, 1\}} \left[W(f) + \frac{(|S(f)| - \sqrt{q})^2}{2q} \right], \quad (50)$$

$$\frac{2M(C)}{(q+1)\sqrt{q}} = 1 - \frac{2}{q+1} \min_f \left[W(f) + \frac{(|S(f)| - \sqrt{q})^2}{2q} \right]. \quad (51)$$

Consequently, along any sequence of such q ,

$$\frac{2M(C)}{(q+1)\sqrt{q}} \rightarrow 1$$

if and only if there are Boolean f_q satisfying

$$|S(f_q)| = o(q), \quad W(f_q) = o(q). \quad (52)$$

Proof. Index C by $\{\infty\} \cup \mathbb{F}_q$ and write

$$C = \begin{pmatrix} 0 & \mathbf{1}^T \\ \mathbf{1} & K \end{pmatrix}, \quad K_{a,b} = \chi(a - b).$$

The Gauss-sum diagonalization says that K has additive-Fourier multiplier 0 at frequency zero and $\epsilon_q \sqrt{q} \chi(r)$ at $r \neq 0$. Therefore Parseval gives

$$f^T K f = \sqrt{q} (E_+(f) - E_-(f)), \quad E_+(f) + E_-(f) = q - \frac{S(f)^2}{q}.$$

For $x = (s, f)$ with $s \in \{-1, 1\}$,

$$Q_C(x) = sS(f) + \frac{\sqrt{q}}{2} (E_+(f) - E_-(f)).$$

Optimizing s gives

$$\max_{s \in \{-1, 1\}} |Q_C(s, f)| = |S(f)| + \frac{\sqrt{q}}{2} \left(q - \frac{S(f)^2}{q} - 2W(f) \right).$$

Subtracting this expression from the spectral ceiling $(q+1)\sqrt{q}/2$ and completing the square proves (50) and (51). The normalized defect tends to zero exactly when the bracket is $o(q)$, which is equivalent to (52). $\square$

Theorem 38 (Least-nonresidue Paley alignment). *Let $p \equiv 1 \pmod{4}$ be prime, let χ_p be the Legendre symbol, and put*

$$\ell(p) = \min\{d \geq 1 : \chi_p(d) = -1\}.$$

Identify $\mathbb{F}_p$ with $\{0, \dots, p-1\}$, set $m = (p-1)/2$, and define

$$f_p(a) = \begin{cases} -1, & 0 \leq a < m, \\ 1, & m \leq a < p. \end{cases}$$

Then $S(f_p) = 1$ and

$$W(f_p) \leq \frac{2p}{\ell(p) - 1}. \quad (53)$$

Consequently

$$\frac{2M(C_{p+1})}{(p+1)\sqrt{p}} \geq 1 - \frac{4p}{(p+1)(\ell(p) - 1)} - \frac{(\sqrt{p} - 1)^2}{p(p+1)}. \quad (54)$$

There is an increasing sequence of primes $p_j \equiv 1 \pmod{4}$ such that

$$\frac{p_{j+1}}{p_j} \longrightarrow 1, \quad \ell(p_j) \longrightarrow \infty.$$

Thus prime-field Paley matrices with nonsquare parameter attain their spectral ceiling asymptotically on a multiplicatively dense sequence:

$$\frac{2M(C_{p_j+1})}{(p_j+1)\sqrt{p_j}} \longrightarrow 1.$$

Proof. Use the standard additive character. For $r \neq 0$, the complete character sum vanishes, so the geometric-sum formula gives

$$|\widehat{f_p}(r)|^2 = \frac{4}{p} \frac{\sin^2(\pi r m / p)}{\sin^2(\pi r / p)}. \quad (55)$$

Writing $\|r\|_p = \min(r, p-r)$ and using $\sin(\pi \|r\|_p / p) \geq 2\|r\|_p / p$ gives

$$|\widehat{f_p}(r)|^2 \leq \frac{p}{\|r\|_p^2}.$$

Here the standard quadratic Gauss sum is $\sqrt{p}$, so the negative Fourier half consists of the nonresidue frequencies. Because $\chi_p(-1) = 1$, every such frequency has $\|r\|_p \geq \ell(p)$. Therefore

$$E_-(f_p) \leq 2p \sum_{d=\ell(p)}^{(p-1)/2} \frac{1}{d^2} \leq \frac{2p}{\ell(p) - 1}.$$

This proves (53); inserting it and $S(f_p) = 1$ in Proposition 37 proves (54).

It remains to construct the prime sequence. For $j \geq 2$, let

$$L_j = 8 \prod_{\substack{\lambda \leq j \\ \lambda \text{ odd prime}}} \lambda.$$

If $p \equiv 1 \pmod{L_j}$, then every integer from 1 through j is a quadratic residue modulo p . Indeed, $p \equiv 1 \pmod{8}$ handles 2; for an odd prime $\lambda \leq j$, quadratic reciprocity gives

$$\left(\frac{\lambda}{p}\right) = \left(\frac{p}{\lambda}\right) = 1;$$

multiplicativity handles composites. Hence $\ell(p) > j$.

For each fixed j , the prime number theorem in the progression $1 \pmod{L_j}$ supplies a threshold X_j beyond which every interval $(x, (1 + 1/j)x]$ contains such a prime [14]. Choose the X_j increasing, and start with a prime $p \geq X_2$ in the progression $1 \pmod{L_2}$. Maintain at level j the invariant $p \geq X_j$. If $p < X_{j+1}$, choose the next prime congruent to 1 $\pmod{L_j}$ in $(p, (1 + 1/j)p]$ and repeat. Once $p \geq X_{j+1}$, choose the next prime congruent to 1 $\pmod{L_{j+1}}$ in $(p, (1 + 1/(j + 1))p]$ and raise the level. Each level is left after finitely many steps because the selected primes are strictly increasing. The levels and hence $\ell(p)$ tend to infinity, while all successive ratios tend to one. $\square$

The interval is a subsequence construction, not a uniform construction over all primes. Along primes $p \equiv 5 \pmod{12}$, the frequencies ± 1 are quadratic residues and ± 3 are nonresidues. For each fixed odd r , (55) gives

$$\frac{|\widehat{f_p}(r)|^2}{p} \longrightarrow \frac{4}{\pi^2 r^2}.$$

The ± 1 pair eventually carries more than half the total energy, whereas the opposite half contains the ± 3 pair. Hence

$$\liminf_{\substack{p \rightarrow \infty \\ p \equiv 5 \pmod{12}}} \frac{W(f_p)}{p} \geq \frac{8}{9\pi^2}. \quad (56)$$

Affine images of the interval have the same obstruction. Thus specially chosen arithmetic progressions, not a pointwise property of intervals, drive Theorem 38.

Proposition 39 (Exact prime-order leakage obstruction). *For prime $p \equiv 1 \pmod{4}$ and nonconstant $f : \mathbb{F}_p \rightarrow \{-1, 1\}$,*

$$W(f) \geq \frac{4(p-1)^2}{p^3} > 0. \quad (57)$$

In particular, $W(f) = 0$ if and only if f is constant.

Proof. Put $P_f(X) = \sum_{a=0}^{p-1} f(a)X^a$. If one Fourier half vanishes, then P_f vanishes at a primitive p th root of unity. Hence $\Phi_p \mid P_f$, and the degree and coefficient constraints force $P_f = \pm \Phi_p$.

For the quantitative statement write $f = 1 - 2\mathbf{1}_A$, replacing f by $-f$ if convenient. Nonconstancy makes A nonempty and proper. If ζ is primitive, then

$$P_f(\zeta) = -2 \sum_{a \in A} \zeta^a.$$

The inner sum is nonzero, since otherwise its 0–1 polynomial would be a multiple of Φ_p , forcing A to be empty or all of $\mathbb{F}_p$. Its algebraic norm is therefore a nonzero integer, and hence

$$|N_{\mathbb{Q}(\zeta)/\mathbb{Q}}(P_f(\zeta))| = 2^{p-1} \left| N_{\mathbb{Q}(\zeta)/\mathbb{Q}} \left(\sum_{a \in A} \zeta^a \right) \right| \geq 2^{p-1}.$$

Put $g = (p-1)/4$. Each quadratic half is closed under conjugation and contains g conjugate pairs. Let $b_1, \dots, b_g$ be the squared moduli for one representative of each pair in the chosen half, and let $c_1, \dots, c_g$ denote the other half. Parseval gives $2 \sum_i c_i \leq p^2$, while AM–GM gives

$$\prod_i c_i \leq \left(\frac{p^2}{2g} \right)^g.$$

The algebraic norm is the product over all $2g$ conjugate-pair squared moduli, so

$$\prod_i b_i \prod_i c_i \geq 2^{p-1} = 16^g.$$

A second AM–GM application therefore gives

$$2 \sum_i b_i \geq 2g \left(\prod_i b_i \right)^{1/g} \geq \frac{4(p-1)^2}{p^2}.$$

Thus

$$\sum_{r \in H_{\pm}} |P_f(\zeta^r)|^2 \geq \frac{4(p-1)^2}{p^2}.$$

Dividing by p gives the stated bound for the chosen half. Repeating the same argument with the two halves interchanged proves it for their minimum $W(f)$ and establishes (57). $\square$

Exact exhaustive evaluation gives

q	5	13	17	29
$M(C_{q+1})$	5	21	33	75
$2M(C_{q+1})/((q+1)\sqrt{q})$	$\sqrt{5}/3$	$3/\sqrt{13}$	$11/(3\sqrt{17})$	$5/\sqrt{29}$

The four displayed ratios increase, ending at $5/\sqrt{29} = 0.9284766908\dots$, but four selected values prove neither monotonicity nor an asymptotic law. The square-order family in Theorem 36 has ratio one identically.

Theorem 38 solves the Fourier character-sum half on a multiplicatively dense prime sequence. It shows only that $M(C_{p_j+1}) \sim (p_j+1)\sqrt{p_j}/2$. Since $F(p_j+1) \leq M(C_{p_j+1})$, this supplies no lower bound on F . To obtain the original limit $1/2$ by this route one would additionally need, along the same sequence,

$$F(p_j+1) \geq (1-o(1))M(C_{p_j+1}). \quad (58)$$

Together with multiplicative density, monotonicity of F , and the existing limsup bound, (58) would force the limit to be $1/2$. The exact optimality at orders 6 and 14 is finite evidence only; order 10 already shows that conference matrices need not be optimal.

The remaining rigidity statement should not be mistaken for a simplification of the asymptotic problem.

Proposition 40 (Dense Paley rigidity is equivalent to the value-1/2 limit). *For the multiplicatively dense sequence $n_j = p_j + 1$ in Theorem 38, the estimate*

$$F(n_j) \geq (1 - o(1))M(C_{n_j})$$

holds if and only if

$$\lim_{n \rightarrow \infty} \frac{F(n)}{n^{3/2}} = \frac{1}{2}.$$

Proof. If rigidity holds, then $F(n_j)/n_j^{3/2} \rightarrow 1/2$. For $n_j \leq n < n_{j+1}$, monotonicity gives

$$\frac{F(n)}{n^{3/2}} \geq \frac{F(n_j)}{n_j^{3/2}} \left(\frac{n_j}{n} \right)^{3/2} = \frac{1}{2} - o(1),$$

because $n_{j+1}/n_j \rightarrow 1$. The global limsup bound supplies the reverse inequality. Conversely, if the full limit is 1/2, then both $F(n_j)/n_j^{3/2}$ and $M(C_{n_j})/n_j^{3/2}$ tend to 1/2, while $F(n_j) \leq M(C_{n_j})$; hence their ratio tends to one. $\square$

Thus the Paley theorem identifies a candidate extremal family and an exact eigenspace mechanism, but the dense-sequence lower bound carries the entire value-specific difficulty of proving the limit 1/2.

17 Seven proof programs and the sharp remaining lemma

The detailed derivations and failed variants are preserved in RESEARCH_LEDGER.md. The current programs are logically distinct.

A. Quenched extensive free energy. Prove $\lim_n \Psi_{n,\beta}$ for every fixed $\beta > 0$, or at least for an unbounded set of β 's. Equation (7) then forces convergence. Theorem 20 gives an exact outer-minimum-compatible recursion, while Theorem 21 closes scalar parameter transport with a leading gap. The surviving state is the conditional relative-switching law in Theorem 23. A useful theorem must derive a variational limit or summable composition defect from this state, rather than rename the desired almost-superadditivity. Proposition 5 separately shows why independent annealed cross blocks fail. A repair must absorb the mixed cross geometry in a joint minimax order parameter, as random-disorder spin-glass interpolation absorbs it only after disorder expectation [3, 4].

B. Near-saturated relaxation. Find a nonconvex compact relaxation whose optimizers have $\sum_e (1 - a_e^2) = o(n^2)$ and an asymptotic composition law. Then Proposition 4 transfers its limit to signs. The cube and sphere calculations prove that the two obvious feasible sets are structurally too loose or exactly discrete.

C. Signed Eulerian polynomial. Prove a second-order limit for $\min_y \log P_y(t)$ after the explicit MacWilliams terms are included. Puncturing and ordinary products fail because of the common parity phase, rank-one cross lift, and macroscopic family of mixed 4-cycles. A quenched coset-weight theorem, rather than an average weight enumerator, is required.

D. Cavity covering profile. Control the Pareto set $(M(B), \delta_w(B))$ uniformly in the near-optimal energy window. Theorem 13 makes this state exact for one vertex, while the order-seven computation rules out the smaller state consisting only of exact maximizers. A useful theorem must either stabilize the normalized dyadic Bellman cost or prove switching-orbit noncoverage for the multivertex rank-one group. Proposition 19 rules out the all-state weighted union bound, so the replacement must exploit intersections among bad constraints. Theorem 26 supplies a concrete intermediate target: prove that the three local deficit profiles cross the 2^{N-1} microcanonical threshold with a power-saving margin. If that scalar target fails, additive labels in the exact convolution remain available. The labeled Parseval bound and complete localizing-moment hierarchy make that refinement exact at every finite order, but the even/odd binomial moment pair rules out bounded-order generic closure. The concrete target is a signing-specific degree- $O(N)$ negative localizing form from the factored shell coefficients with a power-saving, composition-stable error.

E. Second-order limit object. Construct a rate-sensitive topology for $A/\sqrt{n}$ in which the Boolean maximum is continuous and sign arrays have an asymptotically lossless composition. Theorem 10 and Proposition 11 rule out ordinary graphons and empirical spectra. Spectral edges, eigenvectors, or the whole induced Gaussian process would have to remain visible.

F. Paley alignment and minimax rigidity. Theorem 38 proves the Fourier leakage target along a multiplicatively dense sequence of prime Paley orders. The remaining step is the remaining minimax lower bound (58). This is the hard adversarial statement and cannot be inferred from the performance of a selected construction. It would show that every signing, not merely the Paley signing, has asymptotic maximum at least the Paley value. Proposition 40 shows that this dense-sequence rigidity is equivalent to the full value-1/2 conclusion; it is a concrete extremal reformulation, not a reduction in logical strength.

G. Nonexistence. To separate subsequences one needs order-specific lower bounds on $F(n)$, not merely constructions with different energies. Conference, random, and planted matrices supply upper bounds or deliberately inferior examples. The small values fluctuate but do not certify a subsequence mechanism.

No route presently closes. The ultimate finite-temperature lemma remains the following. Scalar PT is no longer a candidate mechanism; the exact alignment chain identifies the nonscalar state a replacement must control.

Quenched minimax free-energy lemma. For every fixed $\beta > 0$, the limit

$$\lim_{n \rightarrow \infty} \min_A \frac{1}{\beta n} \log \sum_{x,s} \exp \left(\frac{\beta s Q_A(x)}{\sqrt{n}} \right)$$

exists.

Proving this lemma for an unbounded collection of inverse temperatures settles the MathOverflow question by (7). In coding coordinates it asks for cancellation in $P_y(\tanh(\beta/\sqrt{n}))$. Any proof that bounds the mixed cross-block contribution independently incurs a leading error; a successful proof must make that contribution part of the optimized variational structure. The next negative-replica target is a controlled hierarchy or variational limit for the relative-switching Hamiltonian beginning with (25). Any truncation must remain uniform at $u = \Theta(n^{-1/2})$ and $q = \Theta(n)$.

Reproducibility and final status

The exact identity checker, exhaustive small-order search, optional Z3 certificate, expected outputs, deterministic seeds, and certificate boundaries are documented in `verification/README.md`. The literature map is in `LITERATURE.md`; the original and second proof-search ledgers are both preserved.

The fixed-density cut-discrepancy equivalence, the adversarial elliptope limit, the code dual and MacWilliams formula, the oriented covariance barrier, the graph-limit obstructions, finite non-heredity, the weighted covering-radius Bellman identity, the exact negative-replica recursion, the scalar transport no-go theorem, the exact conditional-alignment state, the microcanonical profile composition theorem, the quadratic-cumulant profile barrier, the labeled Parseval and complete integer-moment vacancy hierarchy, the fixed-density cross floor, the equal-cloud obstruction, the conference eigenspace formula, the Paley Fourier-leakage identity and its least-nonresidue construction, and the values through order fourteen materially narrow the search. They do not prove existence or nonexistence of the original limit.

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